How Government Debt Shocks Impact the Economy

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Abstract

I identify exogenous government debt shocks using high-frequency movements in US Treasury futures prices around Treasury financing announcements. In a proxy VAR, these shocks result in a surge in inflation and deep reduction in output, even as the monetary base contracts and primary deficits remain stagnant. Yields and bank balance sheets suggest tighter financial conditions. A New Keynesian model augmented with financial intermediaries can generate the joint rise in inflation and declines in output and the monetary base: banks' absorption of new debt tightens firm financing, and higher yields combined with Treasury cash accumulation drain reserves. Under a standard monetary dominance closure, output and inflation responses are far smaller than in the data. Fiscal dominance amplifies both responses and substantially narrows the quantitative gap.

Keywords: government debt supply shocks; high-frequency identification; proxy SVAR; financial intermediation; fiscal dominance; Treasury financing announcements.
JEL Codes: E31; E43; E44; E62; H63; C32.

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1 Introduction

Since 2008, the United States has experienced two historic surges in government debt. Fiscal stimulus during the Great Recession pushed the year-over-year growth rate of debt-to-GDP to roughly 25%. During the COVID-19 pandemic, that rate climbed even higher to roughly 29%. Both events exceed historical precedent. The causal effects of debt itself, however, remain unclear. Debt is typically treated as residually determined by the government budget constraint, and is therefore viewed as a byproduct of spending and taxation. This paper relaxes that assumption, viewing debt as its own fiscal choice variable and studying the causal effects of an unexpected debt issuance independent of tax or spending needs – a *debt shock*.

Using high-frequency fluctuations in US Treasury futures prices around Treasury financing announcements as an instrument in a proxy VAR, I find that a debt shock raises inflation and lowers output even as the monetary base contracts and the primary deficit remains stagnant. This joint response poses a puzzle: how can an economy generate inflation despite the monetary base contracting? Responses of bank balance sheet variables and Treasury yields across maturities suggest that financial conditions tighten in the wake of these debt shocks.

A New Keynesian model augmented with financial intermediaries can qualitatively match the empirical impulse responses. In the model, banks absorb an exogenous increase in government debt. The increase in government debt raises yields, tightening borrowing constraints for firms. Tighter borrowing constraints raise marginal cost, delivering a jump in inflation and a persistent recession. Higher yields and Treasury cash accumulation from the debt issuance drain reserves. Under monetary dominance, the standard nominal closure, a second puzzle emerges: the model's output and inflation responses are far too small relative to the data. Changing the parameterization within reasonable bounds does not solve the magnitude issue. Rather, fiscal dominance brings the model's responses more in line with the empirical responses.

To use the empirical results as a credible benchmark for the model, I must isolate unexpected debt issuance. This isolation is difficult though: debt, as a stock, is slow to adjust, partly anticipated, and co-moves with spending and taxation. High-frequency variation in US Treasury futures prices in a narrow window around Treasury financing announcements accomplishes this isolation. Through Quarterly Refunding Statements and individual auction announcements, new information about debt supply is released to the market at predetermined times. Because these announcements are embargoed, and because futures markets are highly liquid and regularly traded, the new debt sup-

ply information from announcements is incorporated quickly. A larger than expected issuance announcement should lower futures prices, while a smaller than expected issuance announcement should raise futures prices. This identification approach isolates market responses to news about public borrowing, capturing debt shocks that are plausibly unrelated to concurrent macroeconomic developments.

I then aggregate these shocks to a monthly frequency, using them as an external instrument for the reduced-form residual for the 30 year Treasury yield in a Bayesian VAR. This approach allows me to isolate the component of the yield innovation specifically generated by debt shocks. Relevance diagnostics suggest that the proxy is a strong instrument. The identified shock immediately increases outstanding debt and raises interest rates across the yield curve, validating the interpretation that this shock is a debt supply shock.

The responses of the bank balance-sheet variables in the VAR motivate a financial intermediation mechanism, which I capture through a New Keynesian model with financial intermediaries and a working capital constraint. Specifically, after exogenous debt issuance tightens firm borrowing, the amount of labor firms can hire and the amount of capital in which firms can invest decreases as the marginal cost of that labor and investment rises. As a result, output falls. Through the New Keynesian Phillips curve, higher marginal cost increases inflation. The higher interest rates induce households to hold less currency, while the fall in bond prices causes reserves to fall through the central bank's balance sheet.

Under the standard New Keynesian nominal closure – an active Taylor rule and passive fiscal policy – the response on output is roughly two orders of magnitude smaller in the model than in the empirics. The response of year-over-year inflation is 40 times smaller. This quantitative failure makes nominal closure central to the analysis.

Under monetary dominance, the central bank controls inflation through a strong Taylor rule and the federal government adjusts the primary surplus – using lump-sum taxes in my model for parsimony – to stabilize nominal liabilities. With fiscal dominance, the central bank does not set the policy rate aggressively enough to control inflation, and the fiscal authority does not adjust the primary surplus in response to debt. Instead, inflation adjusts to satisfy the intertemporal government budget constraint conditional on the price of bonds. Implementing fiscal dominance in the model delivers a recession twelve times larger and an increase in year-over-year inflation roughly five times larger than the monetary dominance economy. Robustness checks confirm that this amplification is not driven by a particular parameterization, choice of fiscal residual under monetary dominance, or even model environment within the New Keynesian tradition. Fiscal dominance appears

necessary to bring model magnitudes substantially closer to the empirical magnitudes.

Literature Review

My paper is empirically closely related to Mustafi (2024), Phillot (2025), and Bi et al. (2026). These papers use high frequency identification to evaluate government debt shocks in the United States. In many ways, my paper complements the findings in their papers. Bi et al. (2026) find that Treasury supply shocks raise term premia and crowd-out private investment. Phillot (2025) focuses primarily on financial markets, including corporate bonds and stock volatility. Mustafi (2024) is concerned about disentangling maturity shocks from debt volume shocks. My empirical implementation differs from Phillot (2025) and Bi et al. (2026) by using a proxy VAR rather than local projections. Unlike Mustafi (2024), I identify an aggregate debt-supply shock rather than separating debt-level and maturity shocks. More importantly, though, my paper differs in scope. I study the joint monetary-fiscal accounting of these debt shocks and use those responses to evaluate the empirical plausibility of theories of nominal closure.

I use high frequency identification for my empirical strategy, linking my paper to Kanzig (2021), Nakamura and Steinsson (2018), and Gürkaynak et al. (2005). I use the shocks from that high frequency identification as an instrument in a proxy VAR, connecting me to papers such as Mertens and Ravn (2013), Montiel-Olea et al. (2021), Jarocinski and Karadi (2020), and Plagborg-Moller and Wolf (2021). Because I identify a debt shock, my paper similarly relates to the papers empirically identifying other fiscal shocks, such as the tax shocks in Romer and Romer (2010) and the government spending shocks in Ramey (2016).

The price level determination debate connects me to the literature on fiscal dominance. Foundational papers include Sargent and Wallace (1981), Leeper (1991), and Sims (1994). Recent literature includes Leeper and Leith (2016), Cochrane (2023), and Bianchi et al. (2023). I contribute to this literature by using the empirical responses to evaluate the quantitative plausibility of alternative nominal closures. My financial intermediary New Keynesian model connects my paper to Gertler and Kiyotaki (2010), Gertler and Karadi (2011), Carlstrom et al. (2017), Sims and Wu (2021), and Sims et al. (2022). I extend this class of models by explicitly specifying the monetary and fiscal policy blocks needed to compare monetary and fiscal dominance.

The rest of the paper proceeds as follows: Section 2 lays out my empirical identification and results from the proxy VAR. Section 3 describes the New Keynesian model, discusses nominal closure, and presents impulse response functions. Section 4 demonstrates robustness to selected parameter values, nominal closure, and model environment. Sec-

tion 5 concludes.

2 Empirical Analysis

In this section, I describe my identification strategy using high frequency Treasury futures prices and Treasury auction announcements. I report the empirical impulse responses of my shock series using a proxy VAR and empirically explore reduced-form explanations of my shock series.

2.1 Identification

Identifying the impact of government debt on the economy is challenging. Debt issuance is tightly intertwined with fiscal policy – if the government increases its consumption or its social program expenditure (transfers), it needs to increase debt issuance, holding tax collection constant. If tax rates decrease, keeping government consumption constant, debt issuance must increase. The problem is that these scenarios are all *not* debt shocks. They are government spending shocks or tax shocks.[1] A direct debt shock requires the US Treasury to exogenously issue more (or less) debt relative to the government's current expenditure needs and revenue plans. To analyze this shock, I would need access to internal Treasury tax collection projections and maturity analysis and a high frequency tax data series. Alternatively, and more tractably, I estimate a narrower shock: debt issuance unexpected by the market, measured by high frequency changes in Treasury futures prices around Treasury auction announcements.

Auction Announcements

Each time the Treasury seeks to raise funds, it conducts security auctions. These auctions occur hundreds of times a year. In 2025, for example, the Treasury conducted 444 auctions and issued $29.7 trillion in securities (U.S. Department of the Treasury, 2026). Before an auction, the Treasury announces the maturity of the security being auctioned, the amount offered, the auction date, and other relevant information for investors (see appendix A1 for an example of a Treasury announcement). This announcement date is roughly a week before the actual auction. Auction announcements are tightly controlled and embargoed until 11:00 am Eastern on the announcement date.

In addition to auction announcements, the Treasury releases Quarterly Refunding

[1] Much work has been done on these shocks – see Ramey (2016) for example.

Statements (QFRs). QFRs are released at the beginning of each quarter. These statements indicate the maturities and amounts for each maturity the Treasury plans to borrow over the next quarter. QFRs are more "big picture" than individual auction announcements, with the amounts and maturities subject to revision as the quarter progresses. QFRs too are embargoed until a specific time on release day. Up until May 2013, the release time was 9:00 am Eastern, while after May 2013 the release time has been 8:30 am Eastern.

Futures Prices

The Chicago Board of Trade currently offers eight different futures contracts: 2-year, 3-year, 5-year, 10-year, Ultra 10-year, 20-year, 30-year, and Ultra 30-year, though the 3 and 20-year futures are relatively recent (introduced in 2020 and 2022 respectively). Of these, the 2, 5, 10, and 30-year account for almost all of the trading volume, with the 10-year having the highest trading volume (Johnson et al., 2017).

The contract for each maturity exists on the March quarterly cycle (March, June, September, December). To fulfill the contract, Treasuries within a maturity specific time-window must be delivered by the end of the contract month.[2] The futures have a face amount of $200,000 for the 2 and 3-year futures and $100,000 for the other futures, where prices are quoted in points per $2,000 on the 2 and 3-year futures and points per $1,000 on the other futures. For prices over time, see appendix A2.

I use intraday prices from FirstRate Data to get data from 01/01/2008 to 08/23/2024 for the 2, 5, 10, and 30-year contracts. I first find only dates where a Treasury announcement was made, excluding dates where Treasury buyback announcements were made at the same date and time as a Treasury auction announcement.[3] Note that I include cross-maturity auctions in my dataset. For example, if the Treasury announces an auction for a 5-year security, I will include that announcement date for the 2, 5, 10, and 30 year dates. As such, I pick up movements across the yield curve.

Shock Calculation

High frequency identification requires a window around the shock time. Ederington and Lee (1993) find that Treasury futures prices absorb macroeconomic news almost instan-

[2]The 2-year future deliverable, for example, must be a Treasury with a maturity of 1 3/4 to 2 years.

[3]Treasury buybacks occur when the Treasury repurchases US Treasury securities from the market. These are relatively rare, mostly occurring at the very end of my sample with the stated goals of "liquidity support" and "cash management" (Office of Debt Management, 2024) In practice, these remove some off-the-run securities, target specific areas of the yield curve, and can secure more favorable borrowing terms for the federal government. Because buybacks tend to be for more strategic reasons, I exclude them rather than interpret them as negative supply shocks.

taneously: most price adjustment occurs within one minute of an announcement, with volatility remaining significantly elevated for a total of only 15 minutes. Later papers, such as Lin and Qiao (2026) and Benamar et al. (2021) find similarly fast price discovery in Treasury markets. Indeed, recent work using high frequency identification in Treasury futures have used post-shock windows that are 20 minutes, as in Phillot (2025) and Bi et al. (2026), or as short as ten minutes, as in Zhai (2026). The high-frequency monetary policy shock literature uses similarly sized windows. Gürkaynak et al. (2005), Nakamura and Steinsson (2018), and Jarocinski and Karadi (2020) all use 20 minutes as the window after an FOMC announcement. My baseline results use a window of 15 minutes after the release of the Treasury announcement as motivated by Ederington and Lee (1993). In appendix A5, I vary the post-shock window size.

I next need to calculate the size of the announcement shock. To do so, I fit an ARMA (p, q) model to the logged price data 240 minutes before the shock, where p and q minimize the AIC. I then use the ARMA model to predict the log price at the end of the shock window. I then take the difference between the actual price and the predicted price at the end of the window. To ensure I have shocks centered at zero, I demean the resulting series, following Phillot (2025). In practice though, the means are close to zero anyway.

This process leaves me with four individual shock series, one for each maturity. To combine these four separate series into one shock series, I first sum each series by month. I then create weights based on each maturity's share of the total federal debt load. I assign the two-year shock series the portion of federal debt between one and three years, the five year shock series the portion between three and seven years, the 10 year shock series the portion between seven and 15 years, and the 30-year shock series anything over 15 years. This weighting scheme accounts for approximately 70-75% of the federal debt (see appendix A3). I then do a weighted sum. Figure 1 displays the resulting shock series, multiplied by negative one so that a positive shock will be correlated with a positive increase in debt.

First, notice the shape of the shock series. There is a cluster of large positive shocks around June 2009. This cluster of shocks reflects a number of large debt issuances. After the Great Recession, shocks generally decrease in magnitude and hover around zero. In 2020, we see a small positive cluster of shocks followed by a cluster of negative shocks. This result suggests that markets actually expected more deficit-financing during the COVID-19 pandemic than actually occurred.

Given the scale of the pandemic spending, the publicity fiscal packages received, and the fact that large stimulus spending had recently occurred in the Great Recession, it is not unreasonable to suppose that markets expected large amounts of government spending

financed by debt. Lastly, there is a large positive shock late in August of 2023, followed by large negative shocks in November of 2023 and January of 2024. In November 2023, the Treasury released the QFR for Q4. This announcement promised to increase the auction sizes of the two, three, five, and seven year securities. In January 2024, the Treasury signaled that it would need substantially less borrowing than initially projected due to a higher than expected cash balance at the beginning of the new quarter.

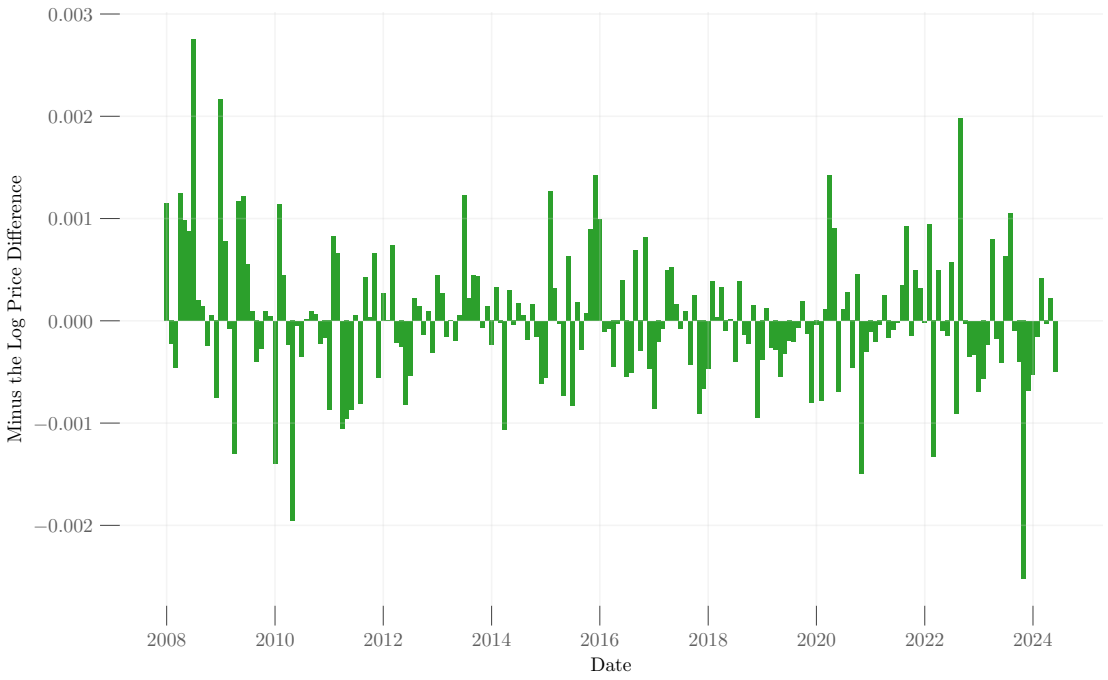


Figure 1: Shock Series

Note: Each green bar denotes a monthly shock, where the y-axis is the negative of the log price difference.

Market Expectations

My identification strategy hinges on the viability of a local version of the efficient markets hypothesis. Namely, I assume that within my calculated windows, Treasury futures markets do not incorporate any new information into the futures price other than the information released in the Treasury auction announcement. I do not require the price level of the futures market to be efficient (Fama (1970) defines an efficient market as one in which "prices always 'fully reflect' available information"). I do require that futures markets price the new information from the auction announcements quickly.

This weaker hypothesis is strongly supported by the literature. Fama (1991) asserts that event studies give supportive evidence for financial markets incorporating new information into prices. Fleming and Remolona (1999) find that secondary Treasury markets

incorporate public macroeconomic news quickly into trading prices. Brandt et al. (2007) demonstrate that Treasury futures markets in particular are important in price discovery of the secondary Treasury market as a whole. In addition, Chordia et al. (2008) show that more liquid markets enhance market efficiency, minimizing arbitrage. Treasury futures markets are highly liquid, with over 3 million contracts traded on an average day in 2017 and low margin requirements enabling active participation (Johnson et al., 2017).

2.2 VAR

I use the following VAR model estimated on monthly data:

$$\boldsymbol{Y}_t = \boldsymbol{\alpha} + \sum_{k=1}^{L} \boldsymbol{A}_k \boldsymbol{Y}_{t-k} + \boldsymbol{u}_t, \qquad \boldsymbol{u}_t \sim \boldsymbol{N}\left(\boldsymbol{0}, \boldsymbol{\Sigma}\right)$$

where $\boldsymbol{Y}_t$ is a vector of endogenous variables at time t, $\boldsymbol{\alpha}$ is a vector of constants, $\boldsymbol{A}_k$ is the matrix of coefficients for the lagged vector of variables for lag $\boldsymbol{k}$, and $\boldsymbol{u}_t$ is a vector of residuals at time t. The residual term is assumed to be normally distributed with variance-covariance matrix $\boldsymbol{\Sigma}$. I employ Bayesian methods in estimation as is standard in the VAR literature (e.g. Jarocinski and Karadi, 2020; Weale and Wieladek, 2016; Banbura et al., 2010). Specifically, I impose a normal-inverse Wishart prior, where the prior mean and variance of the autoregressive coefficients follow a standard Minnesota prior.

In my baseline specification, the endogenous variable vector includes the thirty year Treasury secondary market yield, the log of the S&P 500 price index deflated by the PCE price index, the excess bond premium from Gilchrist and Zakrajsek (2012), year-over-year CPI inflation, the log of the industrial production index, and the first difference of log privately held gross par federal debt deflated by the PCE price index. I use 12 lags, though the results are robust to various lag lengths as seen in appendix A6. The time series runs from 1977M4 through 2024M6.

Since my time series runs through 2024, I include the COVID-19 pandemic in my dataset. Numerous papers have offered solutions to deal with the nonlinearities introduced by the pandemic, including modeling stochastic volatility (Carriero et al., 2024), pandemic indicators (Ng, 2021), nonlinear specifications, (Huber et al., 2023), and dropping pandemic observations (Schorfheide and Song, 2024). I follow Lenza and Primiceri (2022) by modeling volatility during the COVID pandemic as $s_t^2 \boldsymbol{\Sigma}_t$, where $s_t = 1$ prior to March 2020, $\bar{s}_0$ in March 2020, $\bar{s}_1$ in April 2020, and

$$s_t = 1 + (\bar{s}_2 - 1)\rho^{t-2}$$

after April 2020. In effect, this model allows for volatility to jump during the pandemic and smoothly recover according to persistence parameter ρ, preventing the extreme observations during the pandemic from exercising too much control on estimators.

With a high frequency shock series, I can either use the internal instruments approach employed by those such as Romer and Romer (2004) or Jarocinski and Karadi (2020), or the external instruments approach used by Kanzig (2021) and Caldara and Herbst (2019). Because the external instruments approach allows me to estimate the reduced-form VAR over a longer sample than my shock series, and because my shock series covers two unusual periods in economic history – the zero lower bound and COVID-19 pandemic – I follow the external instruments approach. I first estimate the reduced form VAR. Next, I use a Metropolis-Hastings step to update the Lenza and Primiceri (2022) volatility parameters. Then, conditional on the reduced form parameter draws and volatility parameter draws, I estimate a second-step Bayesian regression of the reduced-form residuals on the instrument. This approach propagates the reduced-form posterior uncertainty and the first-stage identification uncertainty while still maintaining the convenience of a Metropolis-within-Gibbs sampler. The cost is that I do not estimate uncertainty in the observed proxy and reduced-form VAR jointly as in Caldara and Herbst (2019).

A valid external instrument, z_t, must satisfy the following two assumptions (Montiel-Olea et al., 2021):

$$\text{Cov}(u_t, z_t) \neq 0 \qquad \text{(Relevance)}$$

$$\text{Cov}(z_t, \varepsilon_{j,t}) = 0 \;\; \forall \; j \neq \ell \qquad \text{(Exogeneity)}$$

where $\varepsilon_{j,t}$ denotes the structural shocks, and ℓ is the index for the targeted structural shock. The relevance condition states that the instrument must be correlated with the reduced-form residuals. The exogeneity condition states that the instrument can only be correlated with one structural shock.

The frequentist literature assesses the relevance condition through tests for weak instruments, such as the robust Wald statistic (Montiel-Olea et al., 2021). I calculate a common-sample robust Wald statistic of 23.8 in my baseline specification, above the threshold of 10 for reliable asymptotic confidence interval coverage.[4] Evidence for exogeneity is more indirect, including testing for the forecastability of the shock series and for correlation with other shock series (Kanzig, 2021). These tests can be found in appendix A4.

[4]Although I use a Bayesian methodology, this statistic is useful as evidence for relevance. The Wald statistic accounting for the full reduced-form sample is 12.3.

My shock series is uncorrelated with other shock series during this time period, shows little evidence of autocorrelation, and is generally not Granger-caused by other relevant variables.

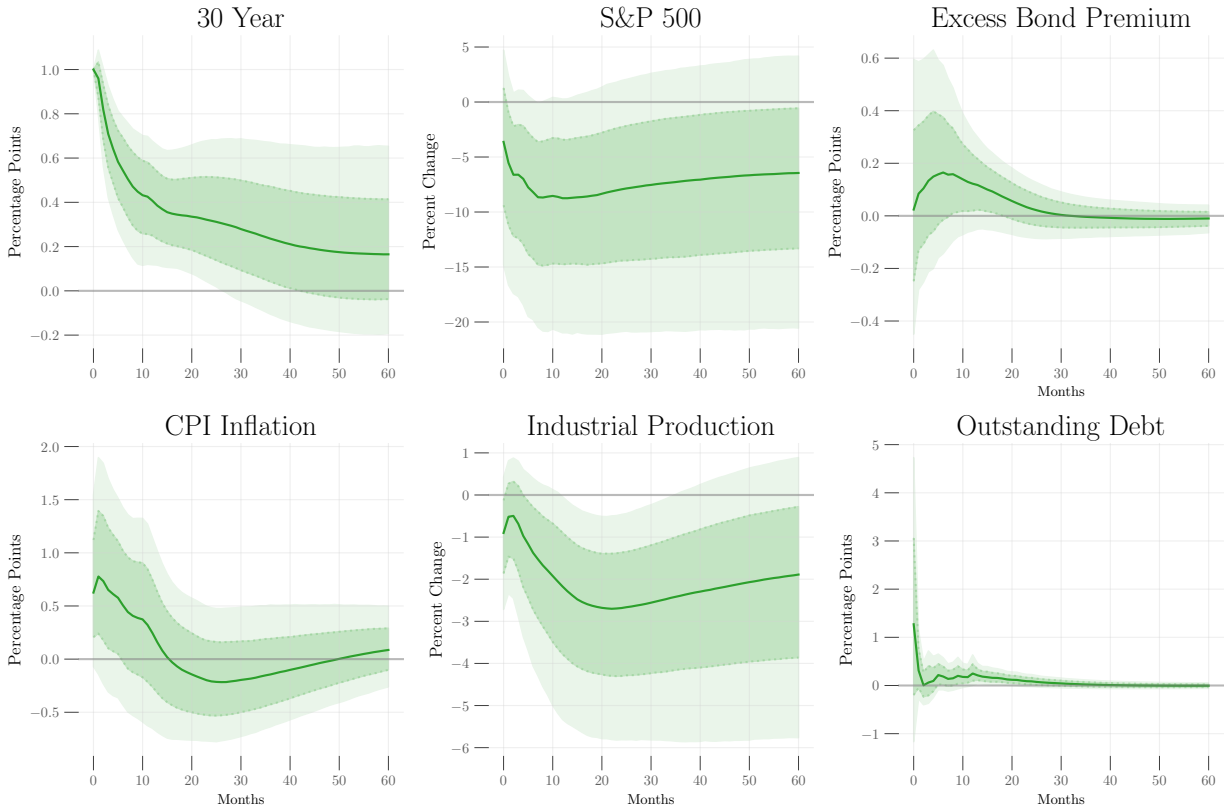


Figure 2: IRFs to an Identified Debt Shock

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set.

Impulse Responses

Figure 2 displays the impulse response functions (IRFs) for a positive shock scaled such that it raises the 30-year yield by one percentage point. The shaded areas denote the 68% and 90% credible sets, respectively. The central task now is to interpret what exactly this shock captures.

First, the identified shock does indeed increase the par amount of debt outstanding. The increase in debt is immediate, with outstanding debt increasing by approximately one percentage point on impact. Debt growth drops to zero in the month following the shock, with perhaps some slow growth about a year later. This result indicates that debt supply is directly impacted by my identified shock.

Second, the movements in the S&P 500 and industrial production preclude this shock from being demand-side stimulation from fiscal policy. The S&P 500 index median response falls on impact by about two percentage points, continues to fall until about one year after the shock to a low of 10 percent, and then slowly recovers over the next four years. The industrial production index does not budge from zero statistically on impact, but declines steadily over the next two years to a median drop of about two-and-a-half percent. After five years, production is still below trend.

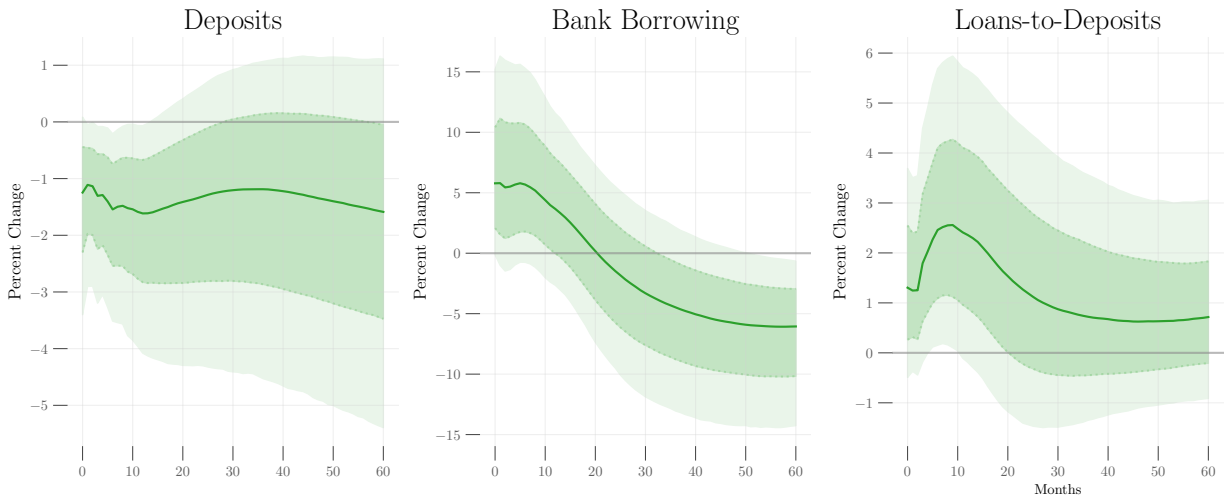


Figure 3: IRFs of Bank Balance Sheet Conditions to an Identified Debt Shock

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set.

Interestingly, year-over-year CPI inflation jumps on impact by about 75 basis points and stays positive for approximately 20 months. After 20 months, a small and statistically insignificant deflationary period sets in until five years after the shock. This result, combined with the decline in output, looks stagflationary. From a standard aggregate demand-aggregate supply perspective, this result indicates a negative supply shock. The excess bond premium weakly suggests that lending conditions have tightened, as the median response increases in a hump-shape to a maximum of roughly 20 basis points ten months after the shock.

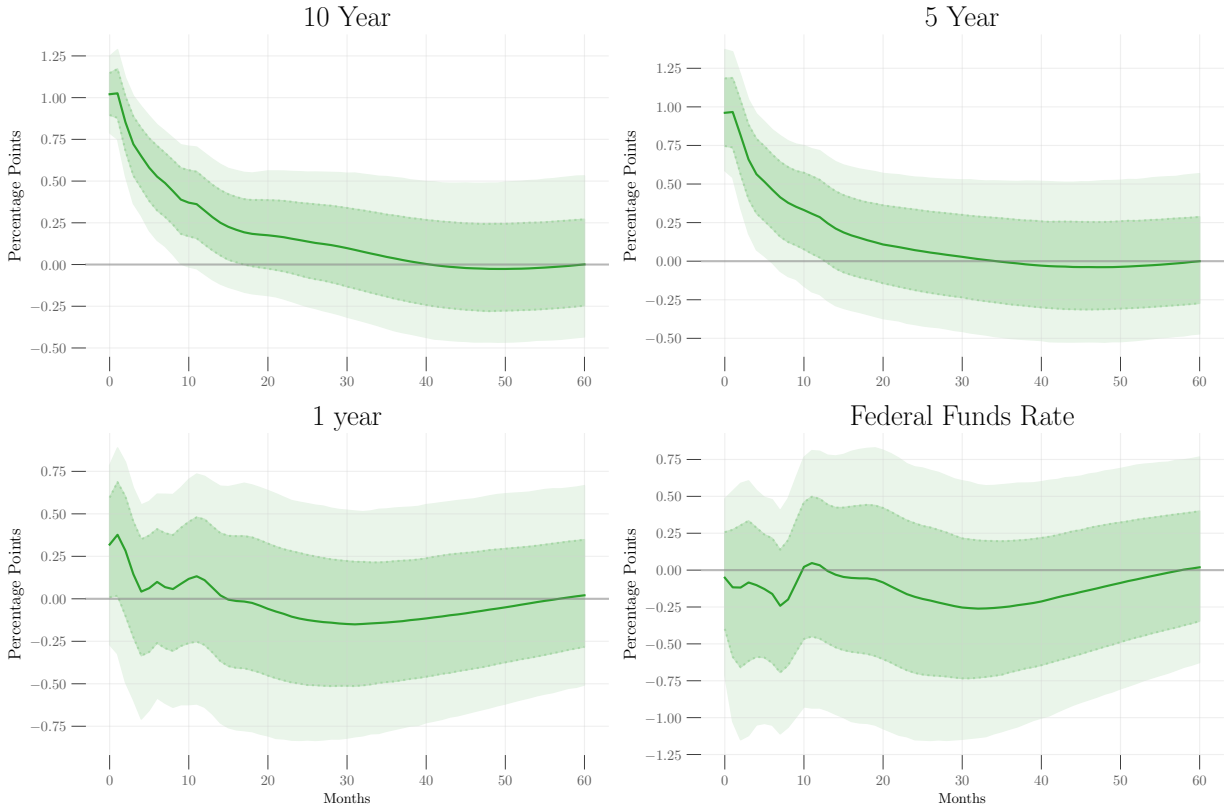


Figure 4: IRFs of Interest Rates to an Identified Debt Shock

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set.

To explore this channel, I add deposits at commercial banks, borrowing by commercial banks, and the loan-to-deposit ratio of commercial banks to my VAR. The results are displayed in figure 3. On impact, deposits fall by approximately one percent – banks are holding less household deposits as collateral. Banks also borrow more. Though the estimate is noisy, the median response indicates that bank borrowing increases by about five percent. Lastly, the ratio of bank lending to deposits increases by roughly two-and-a-half percent ten months after the shock. Taken together, these responses suggest greater reliance on nondeposit funding. A greater reliance by banks on nondeposit funding increases the sensitivity of the financial sector to financial shocks (Cornett et al., 2011). With an increase in the excess bond premium, the impulse responses point toward tightening financial conditions as a result of debt shock.

Maturity or Level?

As discussed in Mustafi (2024), positive debt shocks can be either a level shock – a surprise increase to the stock of debt as a whole – or a maturity shock – a surprise increase in the stock of one particular maturity. To determine which my shock is particularly, I run my baseline VAR with the ten-year, five-year, and one-year secondary market yields, as well as with the federal funds rate. [5] If my shock series was primarily picking up shifts in the Treasury shifting borrowing from one point on the yield curve to another, then the slope or curvature of the yield curve should change. The IRFs can be found in figure 4.

There is some evidence of an increase in the slope of the yield curve. The ten- and five-year secondary market yields jump by approximately one percentage point on impact; this jump is the same size as the constructed one percentage point jump in the thirty-year yield. The one-year yield jumps by only 30 basis points on impact. As such, the medium to long end of the yield curve has increased more than the short-end.

The one-year yield is closely tied to the expected path of the federal funds rate over the near future. As such, monetary policy tends to anchor short-term yields. The federal funds rate does not move at all, so the smaller shift in the one-year yield is to be expected. Beyond the short-end though, yields shift uniformly upward. These responses therefore indicate a strong level shift for most of the yield curve, with a smaller, policy-anchored shift at the short-end.

Reconciling with the Government Budget Constraint

If this shock series is government debt increasing exogenously, then some variable needs to offset that increase in the government budget constraint. A stylized nominal budget constraint is as follows:

$$\Delta B_t = i_{t-1} B_{t-1} + P_t s_t - \Delta(P_t M_t)$$

where B_t denotes government debt, i_{t-1} denotes the nominal interest rate, s_t denotes the primary deficit (government spending minus taxes), M_t denotes money balances, and P_t denotes the price level. If B_t increases, then either $P_t s_t$ must increase or $\Delta P_t M_t$ must fall.

To determine which part of the government budget constraint offsets the increase in debt, I run an additional VAR with deficits-to-GDP and the monetary base-to-industrial production. I construct deficits-to-GDP by taking the quarterly federal government budget surplus or deficit, applying the Census' X13 seasonality filter, and then interpolating

[5] My sample includes the zero lower bound period. I therefore replace the time the federal funds rate is constrained by the lower bound with the shadow rate (Wu and Xia, 2016).

monthly values back to 1977 using the monthly deficit time series following Stock and Watson (2010). I also include a measure of the Treasury's operating cash, also interpolated to monthly values. The IRFs can be found in figure 5.

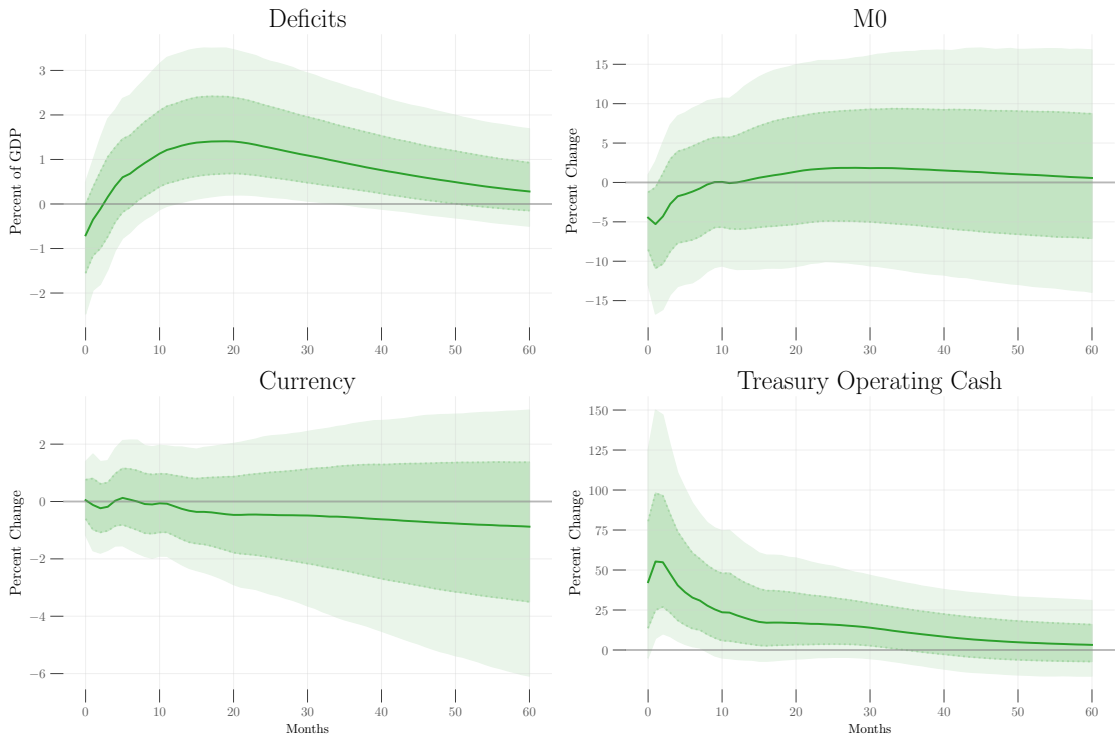


Figure 5: IRFs of Fiscal Variables to an Identified Debt Shock

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set.

The monetary base drops on impact by about five percent before recovering back to zero roughly eight months after the shock. The median response turns positive, reaching a high of about five percentage points two years later, though the response is statistically insignificant. Primary deficits, as a percent of GDP, fall on impact as well, though the response is statistically insignificant, and recover quickly. The small statistically significant increase around 20 months after the shock most likely reflects the decrease in output rather than any increase in the primary deficit itself. Together, these IRFs suggest that the increase in debt does not cause, nor is in response to, an increase in deficits. Rather, this debt shock is more like an asset swap – the Treasury is exchanging debt instruments for liquid money. This perspective is supported by the Treasury's operating cash balance jumping on impact, and returning back to long-run equilibrium over roughly five years.[6]

[6]The Treasury's operating cash balance is highly volatile. That volatility has increased since 2008 (Perli,

This empirical analysis has found that my shock series increases prices and depresses output. The S&P 500 falls. The excess bond premium rises, albeit imprecisely. Bank balance sheet conditions tighten. The yield curve shifts upward, especially on the medium-to-long end. Deficits do not rise on impact, but instead fall. The monetary base contracts and the Treasury's operating cash balance increases.

3 Model

To explain the empirical impulse responses, I follow the literature on New Keynesian models with financial intermediaries (e.g., Gertler and Kiyotaki, 2010; Sims and Wu, 2021). I include money-in-utility to capture currency-in-circulation. I follow Sims et al. (2022) and explicitly model the central bank budget constraint, adding the Treasury General Account as a Treasury asset. To determine the model's nominal anchor, I evaluate the quantitative response of the model to a government debt shock under active monetary policy and active fiscal policy. A full list of equilibrium equations can be found in appendix A7.

3.1 Model Environment

Households

A representative household seeks to maximize:

$$\max_{\{C_t, d_t, M_t^{curr}, L_t\}} \mathbb{E}_0 \left[\sum_{s=0}^{\infty} \beta^s \left(\frac{\left(C_{t+s} - hC_{t+s-1}\right)^{1-\gamma} - 1}{1-\gamma} - \chi \frac{L_{t+s}^{1+\eta}}{1+\eta} + \psi_m \left(1-\rho_m\right)^{\xi} \frac{\left(M_{t+s}^{curr} - \rho_m M_{t+s-1}^{curr}\right)^{1-\xi} - 1}{1-\xi} \right) \right]$$

subject to the real flow budget constraint:

$$C_t + d_t + T_t + M_t^{curr} + \tilde{X}_t = (1 + i_{t-1}^d)\Pi_t^{-1} d_{t-1} + w_t L_t + div_t + \Pi_t^{-1} M_{t-1}^{curr} \tag{1}$$

C_t is consumption, d_t is household deposits held by financial intermediaries, T_t is lump-sum taxes, M_t^{curr} is real currency-in-circulation, $\tilde{X}_t$ are net start-up funds provided by households to financial intermediaries, w_t is real take-home wages, L_t is labor, and div_t is

2025). This underlying volatility, and increase in volatility, contributes to the wide credible sets in the impulse response.

dividends from firms. Π_t is gross inflation. ρ_m imposes habit in currency-in-circulation. The $(1-\rho_m)$ coefficient ensures that ψ_m remains interpretable as a utility scaling coefficient. This specific utility structure delivers an AR(1)-type structure in the first-order condition for currency:

$$\omega_t = \psi_m(1-\rho_m)^{\xi}\left(M_t^{curr} - \rho_m M_{t-1}^{curr}\right)^{-\xi} + \beta\mathbb{E}_t\left[\Pi_{t+1}^{-1}\omega_{t+1} - \rho_m\psi_m(1-\rho_m)^{\xi}\left(M_{t+1}^{curr} - \rho_m M_t^{curr}\right)^{-\xi}\right]$$

where ω_t is the shadow value of wealth. Combining this first order condition with the Euler equation gives:

$$\beta\mathbb{E}_t\left[(1+i_t^d)\Pi_{t+1}^{-1}\omega_{t+1}\right] = \psi_m(1-\rho_m)^{\xi}\left(M_t^{curr} - \rho_m M_{t-1}^{curr}\right)^{-\xi} + \beta\mathbb{E}_t\left[\Pi_{t+1}^{-1}\omega_{t+1} - \rho_m\psi_m(1-\rho_m)^{\xi}\left(M_{t+1}^{curr} - \rho_m M_t^{curr}\right)^{-\xi}\right] \quad (2)$$

Equation 2 links the interest rate on deposits with currency demand. Holding expected inflation and the shadow value of wealth constant, an increase in the interest rate on deposits raises the opportunity cost of holding currency. Habit in currency smooths this adjustment, so currency cannot fully adjust to dampen shocks that impact portfolio allocation.

Labor Markets

Following the differentiated labor structure of Erceg et al. (2000), I incorporate sticky wages through the use of a labor union. Workers provide labor services to a continuum of labor unions, indexed by k, who provide differentiated labor services to a labor packer. The packer is a CES aggregator of all the labor varieties in the economy,

$$L_t = \left(\int_k L_{kt}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dk\right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}.$$

Each union k chooses the nominal sequence of wages to maximize the representative household's utility, accounting for the wages and labor disutility of the household's members, subject to a log-quadratic adjustment cost on nominal wages (Auclert et al., 2024):

$$\max_{\{W_{kt}\}} \sum_{t=0}^{\infty} \beta^t \left(U\left(C_t, M_t^{curr}, L_{kt}\right)) - \frac{\varepsilon_w}{2\psi_w}\ln\left(\frac{W_{kt}}{W_{kt-1}}/\Pi_{ss}\right)^2\right)$$

the household budget constraint, and labor demand $L_{kt} = L_t \left(\frac{W_{kt}}{W_t}\right)^{-\varepsilon_w}$. Union optimization in symmetric equilibrium gives rise to a wage Phillips curve:

$$\ln(\Pi_t^w \Pi_{ss}^{-1}) = \psi_w \left(\chi L_t^{1+\eta} - \frac{\varepsilon_w - 1}{\varepsilon_w} \omega_t w_t L_t\right) + \beta \mathbb{E}_t \left[\ln(\Pi_{t+1}^w \Pi_{ss}^{-1})\right] \tag{3}$$

The wage Phillips curve slows down nominal wage adjustment. In an economic downturn, slower nominal wage adjustment keeps the real wage above the flexible equilibrium wage. Firms demand less labor, depressing hours and output longer.

Production

Production occurs in two stages. A competitive final goods firm aggregates output from intermediate firms, which exist on a unit measure such that each firm $j \in [0, 1]$, using a CES aggregation technology:

$$Y_t = \left(\int_j Y_{jt}^{\frac{\varepsilon_p - 1}{\varepsilon_p}} dj\right)^{\frac{\varepsilon_p}{\varepsilon_p - 1}}$$

Intermediate firms have production function:

$$Y_{jt} = Z_t u_{jt}^{\alpha} K_{jt-1}^{\alpha} \left(L_{jt}\right)^{1-\alpha} \tag{4}$$

where Y_{jt}, u_{jt}, K_{jt}, and L_{jt} denote the intermediate firm's output, capital utilization, capital stock, and labor, respectively. Z_t denotes aggregate productivity. The intermediate firm invests in capital, which evolves according to:

$$K_{jt} = I_{jt} + (1 - \delta(u_{jt}))K_{jt-1} - S(I_{jt}, I_{jt-1})I_{jt} \tag{5}$$

where $\delta(u_{jt})$ denotes a depreciation function in utilization such that $\delta' > 0$ and $\delta'' > 0$, and $S(I_{jt}, I_{jt-1})$ denotes an investment adjustment cost function such that $S(1) = S'(1) = 0$ and $S''(1) > 0$. Each intermediate firm must borrow to finance a portion of its expenses each period:

$$Q_t^L \left(B_{jt}^L - \kappa_L \Pi_t^{-1} B_{jt-1}^L\right) \geq \psi \left(w_t L_{jt} + I_{jt}\right) \tag{6}$$

where Q_t^L denotes the price of firm borrowing, B_{jt}^L denotes the amount of firm borrowing, and κ_L governs the decay rate of bond payments as in Carlstrom et al. (2017). ψ governs the portion of the firm's expenses that must be covered by new borrowing. The firm

takes this borrowing constraint into account when choosing labor and capital each period. When the constraint binds more tightly, the firm demands less labor and capital.

Lastly, the intermediate firm adjusts prices subject to a log-quadratic adjustment cost. Optimization delivers the New Keynesian Phillips curve in symmetric equilibrium:

$$\ln\left(\Pi_t \Pi_{ss}^{-1}\right) = \psi_p \left(mc_t - \frac{\varepsilon_p - 1}{\varepsilon_p}\right) + \mathbb{E}_t\left[\Lambda_{t+1}\frac{Y_{t+1}}{Y_t}\ln\left(\Pi_{t+1}\Pi_{ss}^{-1}\right)\right] \tag{7}$$

where Λ_{t+1} denotes the household's stochastic discount factor.

Financial Intermediaries

Financial intermediaries exist on a unit measure $\ell \in [0,1]$. Each financial intermediary has balance sheet:

$$Q_t^L B_{\ell t}^L + Q_t^B B_{\ell t}^P + M_{\ell t}^{res} = d_{\ell t} + n_{\ell t} \tag{8}$$

where B_t^P denotes the amount of government bonds held by the public, $M_t{}_t^{res}$ denotes the amount of currency reserves, and n_t denotes net worth. Financial intermediaries seek to maximize real net worth:

$$n_{\ell t} = \frac{1}{\Pi_t}\Big[(i_t^B - i_{t-1}^d)Q_{t-1}^B B_{\ell t-1}^P + (i_t^L - i_{t-1}^d)Q_{t-1}^L B_{\ell t-1}^L + (i_{t-1}^r - i_{t-1}^d)M_{\ell t-1}^{res} + (1 + i_{t-1}^d)n_{\ell t-1}\Big] \tag{9}$$

i_t^L and i_t^B denote the implied interest rates on firm borrowing and government borrowing, respectively, i_t^{res} denotes the interest paid on reserves. Financial intermediaries can always store net worth at other intermediaries, so the relevant spread on each asset is the difference between the interest received on that asset and the market deposit rate. Each bank is subject to an incentive compatibility constraint (Sims and Wu, 2021):

$$V_{\ell t} \geq \theta\left(Q_t^L B_{\ell t}^L + \Delta Q_t^B B_{\ell t}^P\right)$$

where $V_{\ell t}$ denotes the intermediary's value function and θ proxies the availability of credit in the financial sector (Sims and Wu, 2021). To ensure an interest rate spread between firm

borrowing and government borrowing, I impose $\Delta \in [0, 1]$. Optimization delivers:

$$\mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_{t+1}^B - i_t^d) \right] = \frac{\mu_t}{1 + \mu_t} \Delta\theta \tag{10}$$

$$\mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_{t+1}^L - i_t^d) \right] = \frac{\mu_t}{1 + \mu_t} \theta \tag{11}$$

$$\mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_t^r - i_t^d) \right] = 0 \tag{12}$$

where μ_t denotes the Lagrange multiplier on the incentive compatibility constraint and Ω_t is an auxiliary optimization variable. Note that equation 12 imposes equality between the reserve rate and the deposit rate.

The banking sector transfers an exogenous increase in government debt held by the public into the real economy. Higher B_t^P tightens the incentive compatibility constraint. Holding deposit rates constant, a tighter constraint increases firm-deposit spreads and government-deposit spreads. Higher firm-deposit spreads tighten the firm borrowing requirement (equation 6).

Central Bank

The central bank holds government bonds, issues reserves, and provides the Treasury General Account. Its balance sheet is:

$$M_t + TGA_t = Q_t^B B_t^{cb} \tag{13}$$

where B_t^{cb} denotes the quantity of government bonds the central bank holds. The monetary base must satisfy:

$$M_t = M_t^{curr} + M_t^{res} \tag{14}$$

The central bank has the following flow budget constraint:

$$\begin{aligned}(1 + \kappa_B Q_t^B) \Pi_t^{-1} B_{t-1}^{cb} + M_t^{curr} + M_t^{res} + TGA_t \\ = \Pi_t^{-1} M_{t-1}^{curr} + (1 + i_{t-1}^{res}) \Pi_t^{-1} M_{t-1}^{res} + \Pi_t^{-1} TGA_{t-1} + Q_t^B B_t^{cb} + T_t^{cb}\end{aligned} \tag{15}$$

where κ_B governs the decay rate of government bond payments and T_t^{cb} denotes central bank remittances to the government.

Government

The real government budget constraint is as follows:

$$Q_t^B B_t + T_t + T_t^{cb} + \Pi_t^{-1} TGA_{t-1} = (1 + \kappa_B Q_t^B)\Pi_t^{-1} B_{t-1} + TGA_t + G_t \tag{16}$$

where G_t denotes government consumption and B_t denotes real government debt. The government issues nominal debt such that real debt evolves according to an exogenous AR(1) rule:

$$\ln(B_t) = (1 - \rho_b)\ln(B_{ss}) + \rho_b\left[\ln(B_{t-1}) + \ln(\Pi_{ss}) - \ln(\Pi_t)\right] + \varepsilon_t^b \tag{17}$$

ε_t^b denotes the government debt shock. Total government debt must always equal debt held by the public and debt held by the central bank.

3.2 Calibration

I calibrate my model to a monthly frequency. I deliberately stay close to parameter values found in other financial New Keynesian models now standard in the literature. A calibration table can be found in appendix A8.

I set steady-state monthly output to unity. I set the monetary base to 20% of annualized output, roughly the average of the monetary base over GDP from 2013 onward. I set currency-in-circulation to 7% of annualized GDP, implying reserves of 13% of annualized GDP. I set the level of the Treasury General Account to 2.5% of annualized GDP, its approximate level from 2021 onward. I set the market value of government-debt-to-GDP to approximately 110% of GDP. The primary surplus, $T - G$, is calibrated to give a steady-state annualized inflation rate of approximately 2%. The interest rate on government bonds is set to 3% annually. The interest rate on firm borrowing is set to 4% annually. The rate on reserves and deposits is set to 2.005% annually, consistent with Sims and Wu (2021).

I set γ, the utility curvature on consumption, and ξ, the utility curvature on currency, to unity. I set η, the inverse Frisch elasticity, to 0.5, consistent with Bigio (2015). Household consumption habit persistence, h, is set to 0.9, consistent with the quarterly estimates of Christiano et al. (2005) and Smets and Wouters (2007). I set habit persistence on currency-in-circulation, ρ_m, to 0.975 to generate gradual adjustment in currency, consistent with evidence of slow adjustment in currency to long-run equilibrium found by Fischer et al. (2004), Deutsche Bundesbank (2009), and Rua (2018).

I set the coupon-decay parameter on government debt, κ_B, such that effective maturity

is 60 months, close to the average remaining maturity of marketable Treasury debt held by private investors (U.S. Department of the Treasury, 2024). I impose the same maturity on firm debt for parsimony. The elasticities of substitution, ε_p and ε_w, are both set to 11. α, the capital share of output, is set to $1/3$ as in Gertler and Karadi (2011). The slopes on the wage and price Phillips curves are both set to $1/20$, consistent with the price Phillips curve found in Born and Pfeifer (2020). Following Sims and Wu (2021), I set ψ, the portion of expenses that firms must borrow, to 0.81. Δ is set to $1/2$ following van der Kwaak (2024).

Finally, for the exogenous government debt rule, I set ρ_b to 0.99. This value implies a half-life of approximately 70 months. Debt expansions are thus persistent, but still stationary in the long-run.

3.3 Monetary and Fiscal Policy Regimes

The price level is jointly determined by monetary and fiscal policy (Leeper, 1991; Sims, 1994; Woodford, 2001; Cochrane, 2023). Leeper (1991) sorts determinate equilibria into two regimes: monetary dominance and fiscal dominance.

Under monetary dominance, the central bank follows a Taylor rule:

$$i_t^{res} = (1-\rho_r) i_{ss}^{res} + \rho_r i_{t-1}^{res} + (1-\rho_r)\phi_\pi \left[\ln(\Pi_t) - \ln(\Pi_{ss})\right] \tag{18}$$

where $\phi_\pi > 1$ ensures the central bank adequately controls inflation. In models where debt endogenously satisfies the government budget constraint, the government must adjust primary surpluses such that the transversality condition on government debt holds. The simplest way to implement this adjustment is to allow lump-sum taxes to adjust, avoiding the real effects of government spending.[7] Monetary dominance is the traditional nominal closure in the New Keynesian literature.

Recently, though, the literature has renewed interest in the fiscally dominant economy. The central bank does not adjust its policy rate to control inflation and the government does not adjust its primary surplus. Rather, the fiscal authority chooses fiscal policy without conditioning its decisions on its budget constraint. Inflation now adjusts to satisfy the intertemporal government budget constraint conditional on bond prices.

Cochrane (2023) makes the point that these two regimes are observationally equivalent from an econometric standpoint. Regressing the policy rate on inflation does not identify the monetary authority's off-equilibrium behavior. As such, empirically constructing the

[7] Allowing government spending to adjust instead would lead to an expansion in spending. An expansion of spending increases output – an opposite result from the empirical IRFs.

Taylor rule will not distinguish between regimes.

My identified debt shocks, however, provide an opportunity to distinguish between regimes. I shock government debt under monetary dominance and then under fiscal dominance while keeping other parameters constant. I then compare the quantitative responses of inflation, output, bond yields, and the money supply between the two policy closures. While not unconditional in the sense of being model-free, this test provides novel insight into how prices are determined in the United States' fiscal-monetary mix.

For the monetary dominance regime, I add an active Taylor rule (equation 18). I set ϕ_π to 1.5 and ρ_r to 0.9, or about 0.7 quarterly, both standard calibrations in the New Keynesian literature. Taxes adjust endogenously to satisfy the government budget constraint. In addition, I posit that Treasury cash management follows the following rule:

$$\ln\left(\frac{TGA_t}{TGA_{ss}}\right) = \rho_{tga} \ln\left(\frac{TGA_{t-1}}{TGA_{ss}}\right) + \gamma_{tga} \frac{Q^B_{ss} B_{ss} \varepsilon^b_t}{TGA_{ss}} \tag{19}$$

This rule suggests that the Treasury maintains a portion of surprise bond issuances, γ_{tga}, in cash. I set γ_{tga} to 0.65 to match the observed jump on shock impact in the TGA from the empirical IRFs. I set ρ_{tga} to 0.98 to generate a persistent decay in the General Account, as is seen in the empirical IRFs. With fiscal dominance, taxes no longer adjust, though the interest rate on reserves will adjust through equation 12 as the deposit rate adjusts to satisfy equation 2.

3.4 Impulse Response Functions

I shock ε^b_t to simulate a 1.25 percent exogenous increase in government debt, as seen in the empirical IRFs. The innovation has no persistence. I start with the monetary dominance economy and display the resulting impulse response functions in figure 6. Recall that the goal of this exercise is not to exactly match the empirical responses, which would require an IRF matching procedure, but rather to evaluate the model's performance under standard calibrations with different monetary/fiscal regimes.

In the first plot, note first that output falls persistently. This recession occurs due to a fall in labor hours and a fall in investment, which in turn are caused by the drop in the market value of firm loans. The increase in government debt causes interest rates to rise across all forms of borrowing due to the financial intermediaries' optimality conditions. Higher interest rates decrease the market value of the firm's loan. As such, the firm borrowing constraint (equation 6) binds tighter after the debt shock, increasing firm marginal cost. Higher marginal cost lowers the amount of investment and labor the firm

can employ, and, through the New Keynesian Phillips curve, raises prices.

Because the government bond yield increases, the price of government bonds falls. Falling government bond prices cause a reduction in monetary reserves through the central bank balance sheet (equation 13). For parsimony, I do not specify a rule for the central bank to adjust its government bond holdings. As such, the money supply drops as government bond yields rise. Reserves fall immediately, as currency-in-circulation is sluggish to respond.

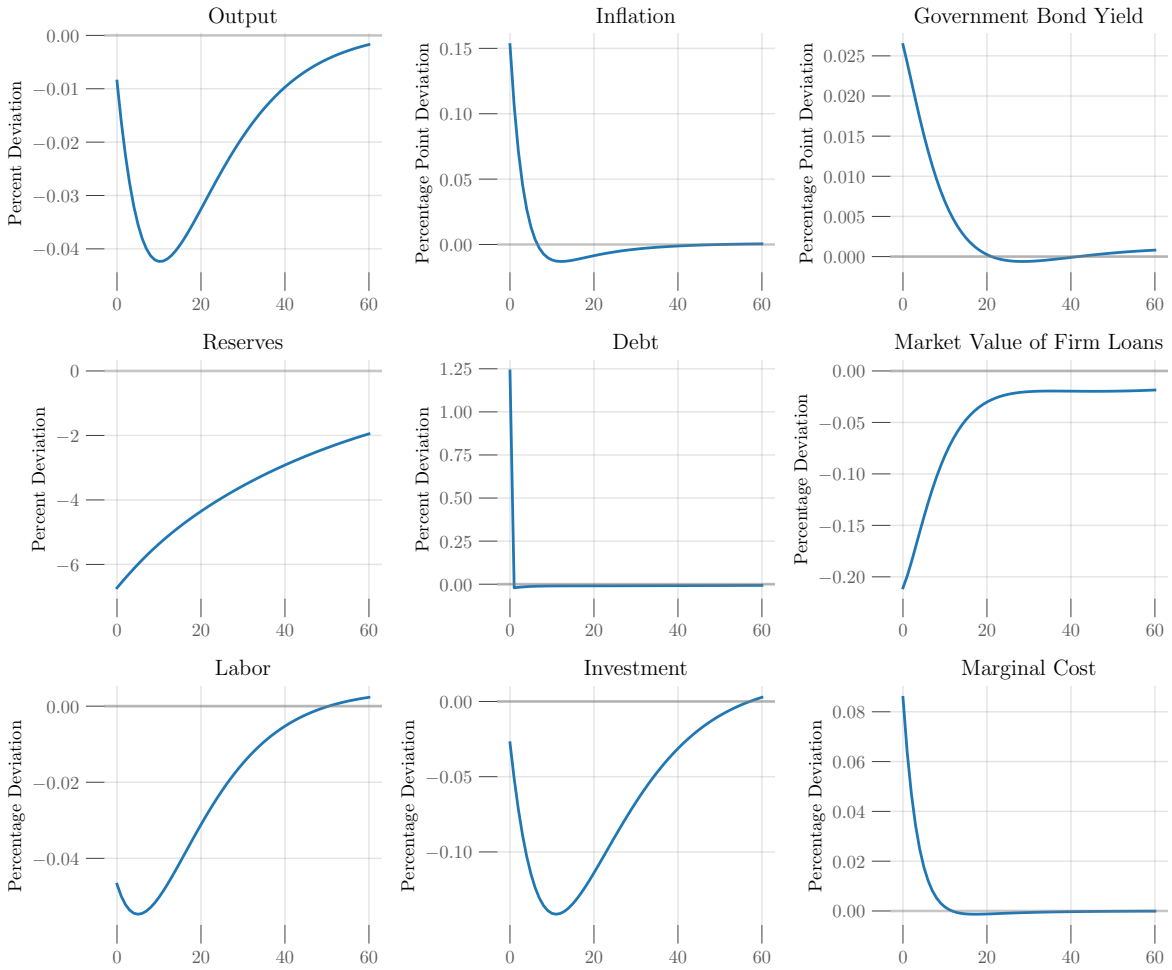


Figure 6: Impulse response functions under monetary dominance to a 1.25 percent exogenous increase in government debt.

Importantly, the monetary dominance model gets the qualitative responses correct for each variable. Output falls, inflation rises, the money supply falls, and interest rates rise. The mechanism of firm-loan crowd-out therefore generates the right directions for the model economy. The quantitative response, though, is not particularly close to the empirical results. Output falls by between one to four percent empirically, while median

year-over-year inflation rises by 75 basis points. The model only generates a 0.04 percent drop in output and only a 0.15 percentage point increase in annualized monthly inflation on impact. The output response is too small by a factor of at least 20, while the impact response in inflation is too small to generate a 75 basis point year-over-year inflation rate.

I now turn to the fiscal dominance economy. Recall that the fiscal dominance economy removes the Taylor rule as an equilibrium equation. The impulse responses to a debt shock under fiscal dominance are found in figure 7.

First, notice that swapping from monetary dominance to fiscal dominance does not significantly change the qualitative response of any variable – the mechanism remains intact. Quantitatively, though, the response of the model is much closer to the empirical findings.

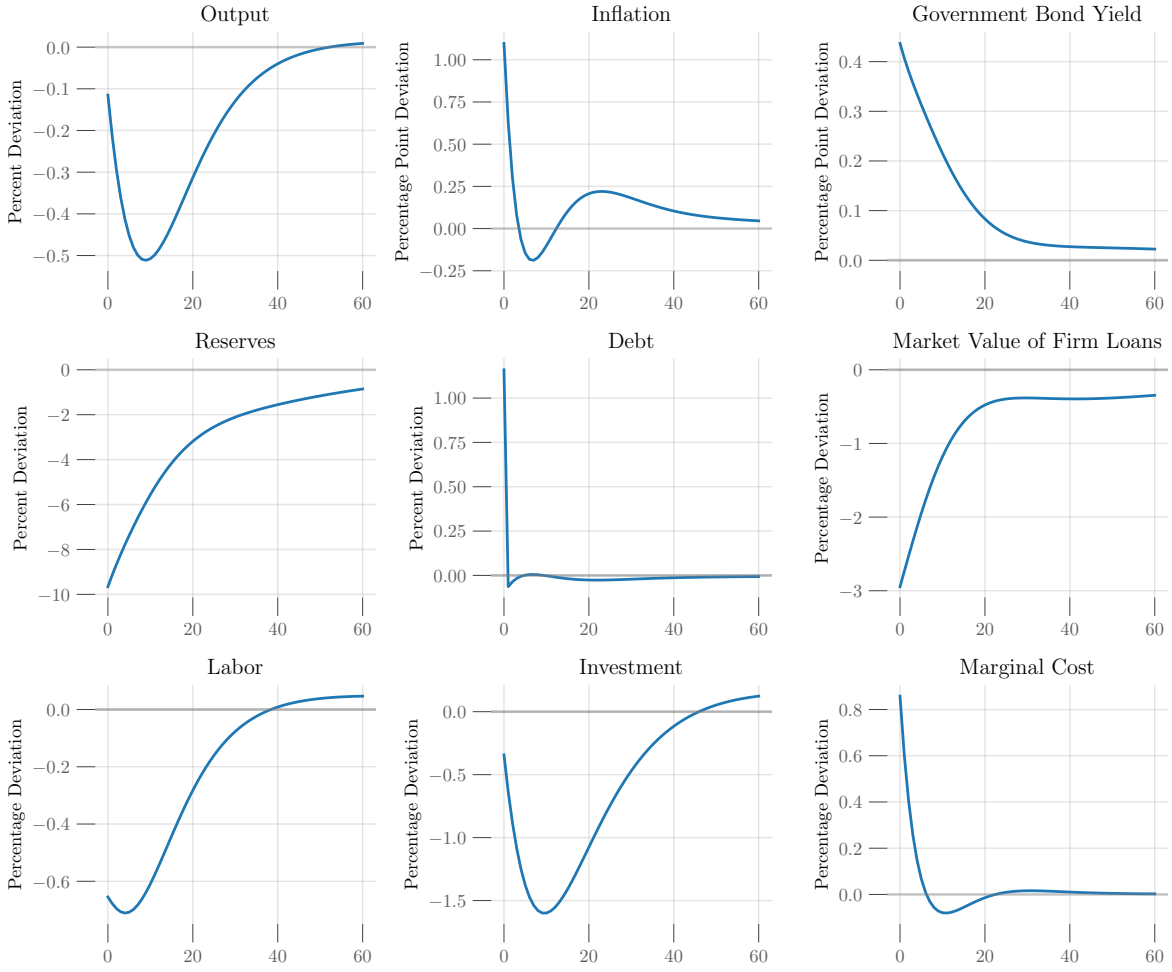


Figure 7: Impulse response functions under fiscal dominance to a 1.25 percent exogenous increase in government debt

The impact response of inflation, relative to the monetary dominant economy, jumps one percentage point on impact. Outside of a small second increase in inflation about 15 months after the shock, inflation then returns back to trend. This inflationary result is the direct cause of fiscal dominance – as nominal liabilities increase for the government, neither taxes nor government spending adjust to ensure that the discounted sum of future surpluses offsets the increase in nominal liabilities. As such only prices can adjust to balance the government budget constraint, leading to a surge in inflation.

Because financial intermediaries take into account future inflation in pricing interest spreads, this surge in inflation raises interest rates more than under monetary dominance. The government bond yield, for example, jumps by approximately 45 basis points and then declines, as opposed to an initial jump of only 2.5 basis points under monetary dominance. The effect on firms is amplified – as bond prices drop the market value of firm loans drops 15 times more than under monetary dominance. The borrowing constraint on the firm now binds much tighter, causing marginal cost to jump by almost 1 percent, which causes both labor and investment to fall by 0.6 and 1.5 percent respectively. Output falls by a respectable 0.5 percent.

Reserves fall by about 9 percent on impact, while currency-in-circulation smoothly declines to a low of minus three percent before slowly recovering. Together, these deliver an approximate six percent decrease in the monetary base – in line with the initial empirical impulse response.

Overall, the fiscal dominance model can deliver both the inflationary surge and steep output drop observed in the empirical IRFs. The increase in the bond yield, while still smaller than the empirical response, approaches the magnitudes seen in the VAR.

4 Model Robustness

The New Keynesian financial intermediary model is a reasonable model to employ in the scenario studied in this paper, given the responses of the banking variables in Section 2.2. In addition, I deliberately maintained a calibration of the model close to established literature values. In this section, I test model robustness. First, I vary key parameters in the model to study whether the quantitative magnitudes of fiscal dominance are dependent upon the specific parameterization I chose. Second, I add a labor income tax to the model as another potential nominal closure for the monetary dominance economy. Lastly, I simplify the model to simple New Keynesian money-in-utility model to test whether the main mechanism – debt induced monetary drain – can deliver similar IRFs without the financial constraints in the baseline model.

Robustness to Parameterization

The first parameters to vary are those not established by the literature, namely ρ_{tga} and γ_{tga}. The values for these parameters were selected to deliver an empirically plausible path for the Treasury General Account in the model. In addition, I vary the share of firm expenses that must be funded by newly issued debt, ψ, the strength of investment adjustment costs, κ_I, currency-in-circulation smoothing ρ_m, bank leverage, ϕ, the Phillips curve slopes, ψ_p and ψ_w, and the Taylor rule weight on inflation, ϕ_π. Each of these parameters controls the pass-through of an increase in government debt to output and inflation.

In Appendix A9, I display the full paths of output and inflation under each regime for the parameter changes. For every variation, fiscal dominance generates a deeper recession and a larger initial inflationary surge than monetary dominance. Indeed, most parameters do not change the magnitudes of either inflation or output much under either nominal closure. The notable exceptions are the TGA parameters and investment adjustment costs under fiscal dominance.

TGA persistence is particularly important for the timing of inflation. The less persistent the value of cash in the TGA, the smaller the initial inflationary surge under fiscal dominance and the larger a secondary inflationary hump. The TGA rule governs the timing of nominal liabilities, which changes the path of interest rates and bond prices. As in Cochrane (2023), this timing change can result in a delayed inflationary surge.

The TGA cash share similarly matters for the initial inflationary spike. The larger the TGA cash share (how much of the debt shock maps directly into saved Treasury cash), the lower the nominal liability increase for the government, and the less inflation is needed to balance the intertemporal government budget constraint. Lower inflation limits the fall in real bond values, resulting in less restrictions on the firms' ability to borrow and alleviating downward pressure on output.

Investment adjustment costs primarily impact the depth and persistence of the recession. Lower adjustment costs allow investment to fall more quickly, deepening the recession. But lower adjustment costs also allow investment to recover more quickly, leading to a less prolonged recession.

Income Tax Rate

I now augment the baseline model with a labor income tax, τ_t^ℓ. The wage Phillips curve becomes:

$$\ln(\Pi_t^w \Pi_{ss}^{-1}) = \psi_w \left(\chi L_t^{1+\eta} - \frac{\varepsilon_w - 1}{\varepsilon_w} \omega_t (1 - \tau_t^\ell) w_t L_t \right) + \beta \mathbb{E}_t \left[\ln(\Pi_{t+1}^w \Pi_{ss}^{-1}) \right] \tag{20}$$

The Treasury's budget constraint adds an income tax term:

$$Q_t^B B_t + T_t + \tau_t^\ell w_t L_t + T_t^{cb} + \Pi_t^{-1} TGA_{t-1} = (1 + \kappa_B Q_t^B)\Pi_t^{-1} B_{t-1} + TGA_t + G_t \qquad (21)$$

I set the steady-state income tax rate to 20% and solve for the lump-sum tax that leads to an annualized steady-state inflation of 2%. Under monetary dominance, τ_t^ℓ will now serve as the residual for the Treasury budget constraint. The resulting IRFs to a 1.25% debt shock for output and inflation are found in figure 8.

Monetary dominance with an income tax closure flips the sign on output and inflation. Output now rises in a hump-shaped pattern for 30 months, inflation falls for the first six months. Neither response qualitatively matches the empirical IRFs. This response occurs as an exogenous increase in debt necessarily leads to a drop in tax revenue. A drop in tax revenue occurs due to a drop in the tax rate.

Fiscal dominance, on the other hand, still delivers the deep and persistent recession and the initial surge in inflation. While the income tax does not close the model with fiscal dominance, endogenous labor supply and wage rates change the amount and timing of nominal liabilities for the government. As a result, output falls slightly less than before, to a trough of roughly -0.45%. Inflation becomes more front-loaded, with an initial increase of roughly 1.75 percentage points.

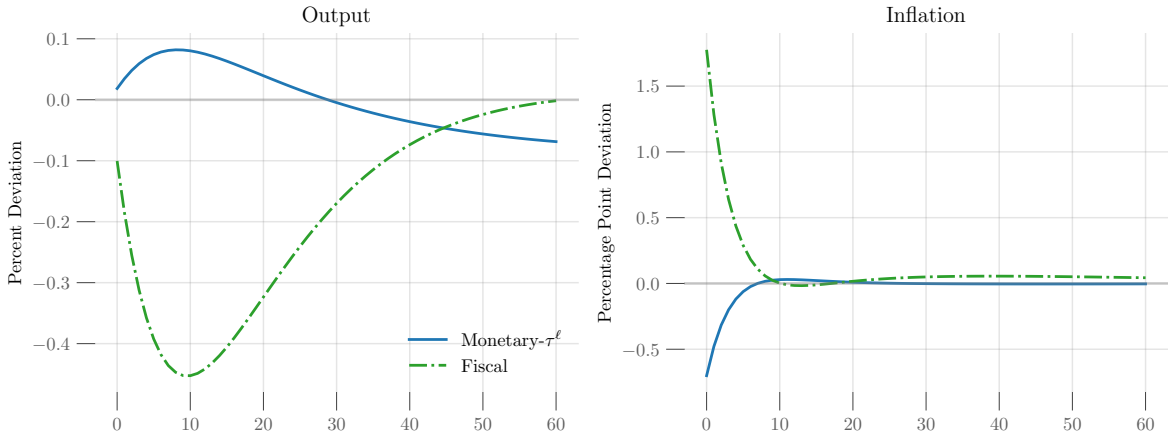


Figure 8: Impulse Response Functions to a 1.25 percent exogenous increase in government debt for a model with income taxes.

Simple Money-in-Utility Model

Rather than including a financial sector, separate Treasury and central bank balance sheets, and firm borrowing, I posit a textbook New Keynesian model with money-in-utility. Un-

der fiscal dominance, money acts as the residual for the consolidated government budget constraint. Under monetary dominance, I allow either government spending or lump-sum taxes to adjust to serve as the residual. Firms still face Rotemberg adjustment costs, but wages are now flexible. A list of equilibrium equations can be found in Appendix A10. In figure 9, I plot output and inflation under the three possible nominal closures.

In the standard New Keynesian model, government debt is Ricardian equivalent. Therefore, if an expansion of debt causes a drop in lump-sum taxes, then the macroeconomy is unaffected. If the expansion of debt causes an increase in government spending, then the model essentially creates a government spending shock – both inflation and output jump. The increase in output is incorrect according to the empirical IRFs though.[8]

Under fiscal dominance, the model delivers the correct qualitative signs. Output falls and inflation rises. Admittedly, the magnitude in the fall in output is far too small and does not persist. Correct impact signs, however, are still an important result. This model does not feature a financial constraint through which government debt can restrict production and raise prices. Instead the contraction in output occurs due to household demand for currency. As currency falls in response to debt, interest rates must rise. Increasing interest rates induces intertemporal substitution through the Euler equation.

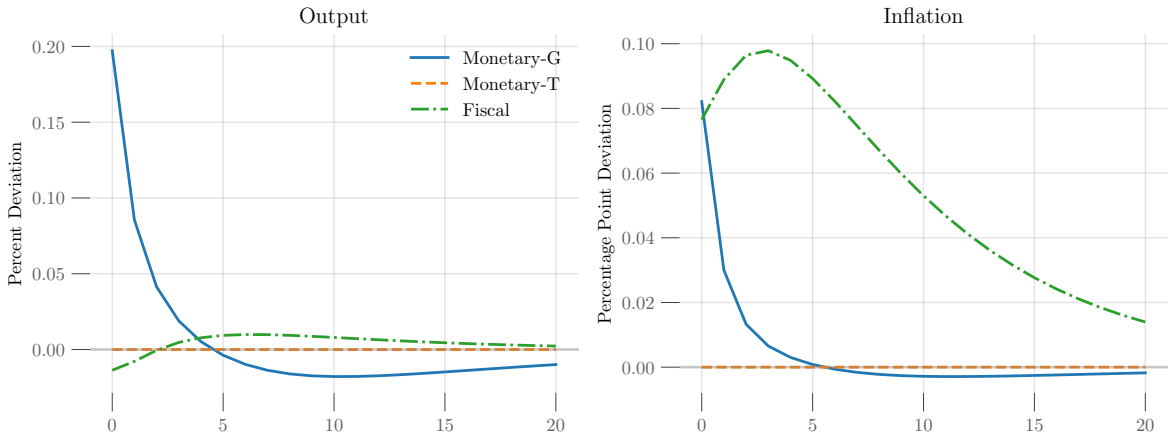


Figure 9: Impulse Response Functions to a 1.25 percent exogenous increase in government debt for a textbook money-in-utility model.

[8]Another possible closure is a shock to government transfers. In a TANK model with hand-to-mouth households, transfers act like expansionary fiscal policy and thus deliver increasing output and inflation.

5 Conclusion

In this paper, I empirically identify government debt supply shocks using high-frequency variation in Treasury futures markets around Treasury financing announcements. I then use a proxy VAR to study the macroeconomic implications of these shocks. A positive debt supply shock tightens financial conditions, decreasing output and increasing inflation. Simultaneously, the monetary base contracts while the primary deficit remains statistically unchanged.

A New Keynesian model augmented with financial intermediaries, a working capital constraint, and explicit central bank and Treasury balance sheets can qualitatively deliver the empirical impulse response functions. Under monetary dominance, though, the quantitative magnitudes are far too small. To shrink the quantitative gap, fiscal dominance is necessary. This result holds across various parameter combinations and within a model with income tax rates as a nominal closure for monetary dominance. Further, in a textbook New Keynesian model augmented with money-in-utility only, fiscal dominance is necessary even for qualitatively correct impulse response functions. Future work should focus on the financial plumbing behind government debt issuance and on the implications of fiscal dominance on the transmission of fiscal policy.

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A Appendix

A1 Treasury Announcement

TREASURY NEWS

Department of the Treasury • Bureau of the Fiscal Service

Embargoed Until 11:00 A.M.
May 14, 2024

CONTACT: Treasury Auctions
202-504-3550

TREASURY OFFERING ANNOUNCEMENT [1]

Term and Type of Security	28-Day Bill
Security Description	4-Week Bill
Offering Amount	$80,000,000,000
Currently Outstanding	$135,590,000,000
CUSIP Number	912797KF3
Auction Date	May 16, 2024
Original Issue Date	February 20, 2024
Issue Date	May 21, 2024
Maturity Date	June 18, 2024
Maximum Award	$28,000,000,000
Maximum Recognized Bid at a Single Rate	$28,000,000,000
NLP Reporting Threshold	$28,000,000,000
NLP Exclusion Amount	$47,300,000,000
Minimum Bid Amount and Multiples	$100
Competitive Bid Rate Increments [2]	0.005%
Maximum Noncompetitive Award	$10,000,000
Eligible for Holding by Treasury Retail	Yes
Estimated Amount of Maturing Bills Held by the Public	$212,949,000,000
Maturing Date	May 21, 2024
SOMA Holdings Maturing	$674,000,000
SOMA Amounts Included in Offering Amount	No
FIMA Amounts Included in Offering Amount [3]	Yes
Noncompetitive Closing Time	11:00 a.m. ET
Competitive Closing Time	11:30 a.m. ET

[1]Governed by the Terms and Conditions set forth in The Uniform Offering Circular for the Sale and Issue of Marketable Book-Entry Treasury Bills, Notes, and Bonds (31 CFR Part 356, as amended), and this offering announcement.

[2]Must be expressed as a discount rate with three decimals in increments of 0.005%, e.g., 7.100%, 7.105%.

[3]FIMA up to $2,000 million in noncompetitive bids from Foreign and International Monetary Authority not to exceed $500 million per account.

A2 Treasury Futures Prices

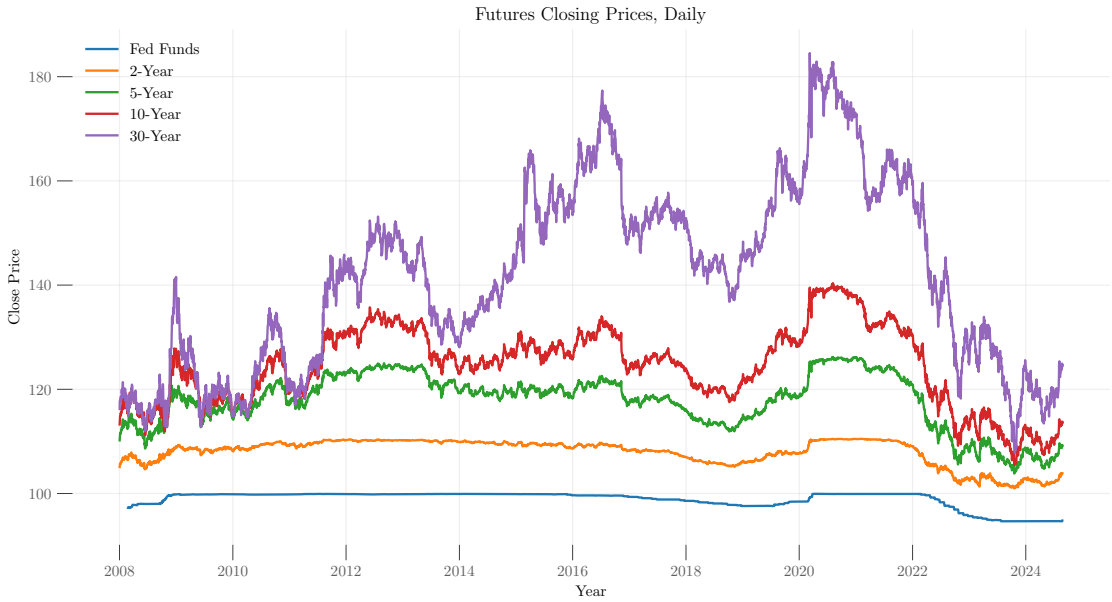


Figure A1: Evolution of Treasury futures prices from 2008 to 2024.

A3 Weighting Scheme

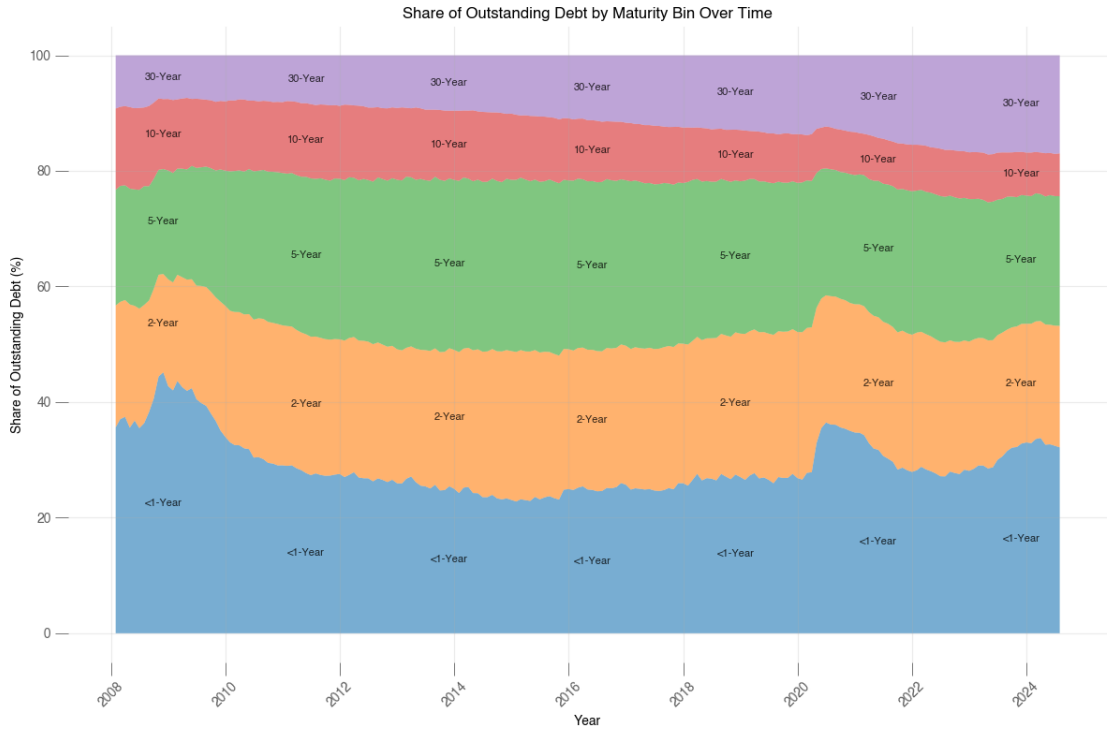


Figure A2: Proportion of outstanding debt by maturity bin.

A4 Diagnostic Tests

Autocorrelation

The first diagnostic test I run is an autocorrelation test. Ideally, a shock should not be autocorrelated with itself. As shown by the autocorrelation plot below, my shock series is not autocorrelated. Only one lag length, 14, is statistically significant out of 20 tested lag lengths.

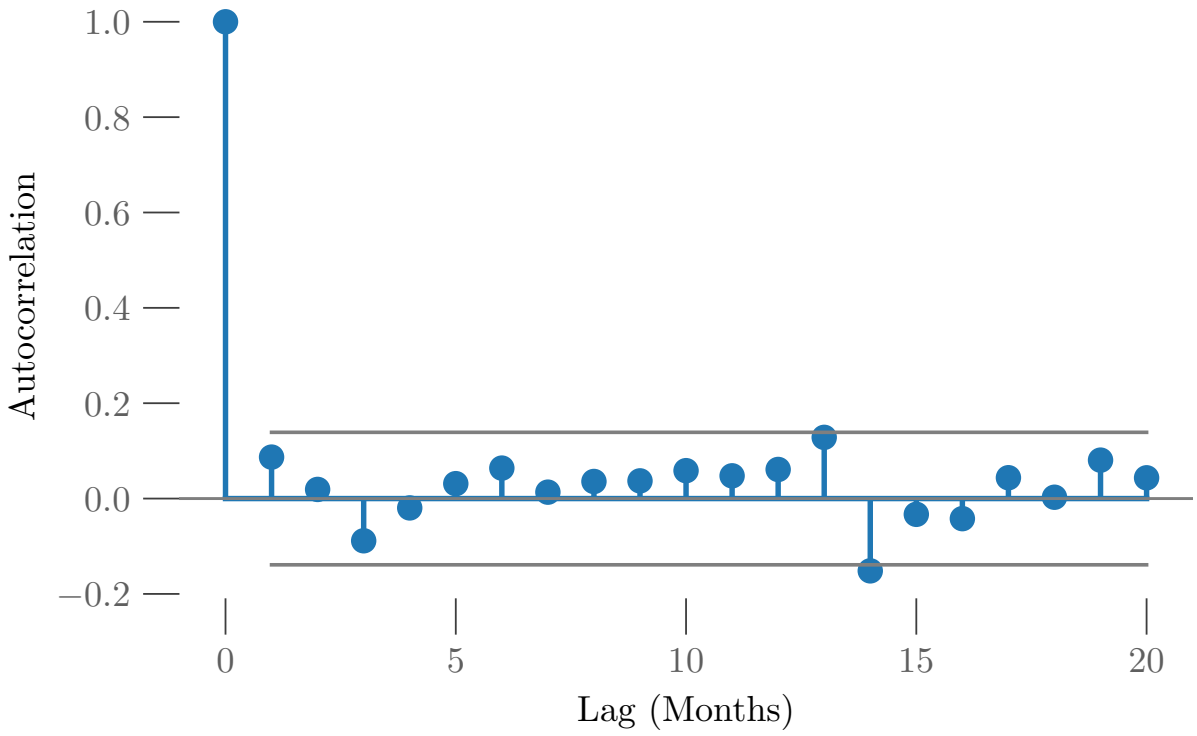


Figure A3: Debt supply shock series correlation with lags of itself.

Correlation with Other Shocks

I take other shock series that run at least part-way through my shock series and calculate the correlation coefficient with my shock. Table A1 summarizes the results. No shock series is statistically correlated with my debt shock series.

Table A1
Correlation Analysis of Shock Series

Shock Series	Type of Shock	Corr.	p-value	Time Period
Nakamura and Steinsson (2018)	Monetary Policy	-0.01	0.91	2008M1-2024M7
Gürkaynak et al. (2005) Target	Monetary Policy	-0.09	0.22	2008M1-2024M7
Gürkaynak et al. (2005) Path	Monetary Policy	0.06	0.42	2008M1-2024M7
Ramey (2016)	Gov. Spending News	0.11	0.62	2008Q1-2013Q4
Gilchrist and Zakrajsek (2012)	Financial	-0.05	0.66	2008M1-2015M12
Baker et al. (2016)	Uncertainty	0.03	0.77	2008M1-2017M12

Note: Correlation between my debt shock series and other relevant shock series. The monetary policy shock series are taken from Acosta et al. (2024). The Ramey (2016) government spending news shock series comes from Ramey's website. Gilchrist and Zakrajsek (2012) and Baker et al. (2016) both come from Kanzig (2021).

Granger Causality

I conduct Granger causality tests between my shock series and a number of macro-financial time series. Table A2 displays the estimates. The results here generally support the exogeneity of my shock series. Over the time-period, only the change in the unemployment rate is significant at the ten percent level. Under the joint null of no predictability, the expected number of false positives at the 10% level across ten tests is one.

Table A2
Granger Causality Tests

Time Series	F-stat	p-value	Time Period
5 Year Yield	0.97	0.48	2008M1-2024M7
10 Year Yield	0.70	0.75	2008M1-2024M7
30 Year Yield	1.20	0.29	2008M1-2024M7
Fed Funds Rate	1.50	0.13	2008M1-2024M7
PCE Inflation	1.34	0.20	2008M1-2024M7
Unemployment	1.73	0.07	2008M1-2024M7
VIX	1.17	0.31	2008M1-2024M7
Deficit	0.98	0.47	2008M1-2024M7
Monetary Base (M0)	1.21	0.28	2008M1-2024M7
Industrial Production	1.16	0.32	2008M1-2024M7

Note: Granger causality tests of various macro-financial time series on my shock series. Each test includes a constant and 12 lags of the macro-financial series and the shock series. I first-difference each macro-financial series, with the exception of year-over-year inflation. During the zero lower-bound period, I use the shadow rate of Wu and Xia (2016). Lenza and Primiceri (2022) find that for parameter estimation, dropping COVID observations is acceptable. As such, I drop the COVID months of March-May 2020 and the lags that depend on those months. I also include an intercept shift for data after March 2020. Joint F-statistics are calculated using a Newey-West covariance estimator.

A5 Robustness to Window Choice

In my baseline specification, I choose 15 minute windows after the announcement as motivated by the literature on high frequency identification. Here, I vary the windows. I first double the window to 30 minutes. I then expand to 60 minutes as a wide stress test. Of course, longer windows potentially allow more contamination into the calculated shock. The impulse responses, though, are mostly stable. Industrial production still falls and median inflation, at least initially, still rises. Inflation does move negative more quickly, albeit imprecisely, than with the 15 minute windows. In addition, the longer the window, the more the excess bond premium jumps on impact rather than rising in a hump-shape.

Lastly, I include a short five minute window specification. The inflation surge becomes stronger, and industrial production falls by more. The response of outstanding debt, though, becomes imprecise, with the 90% credible set including debt growth up to two percentage points or down to minus four percentage points.

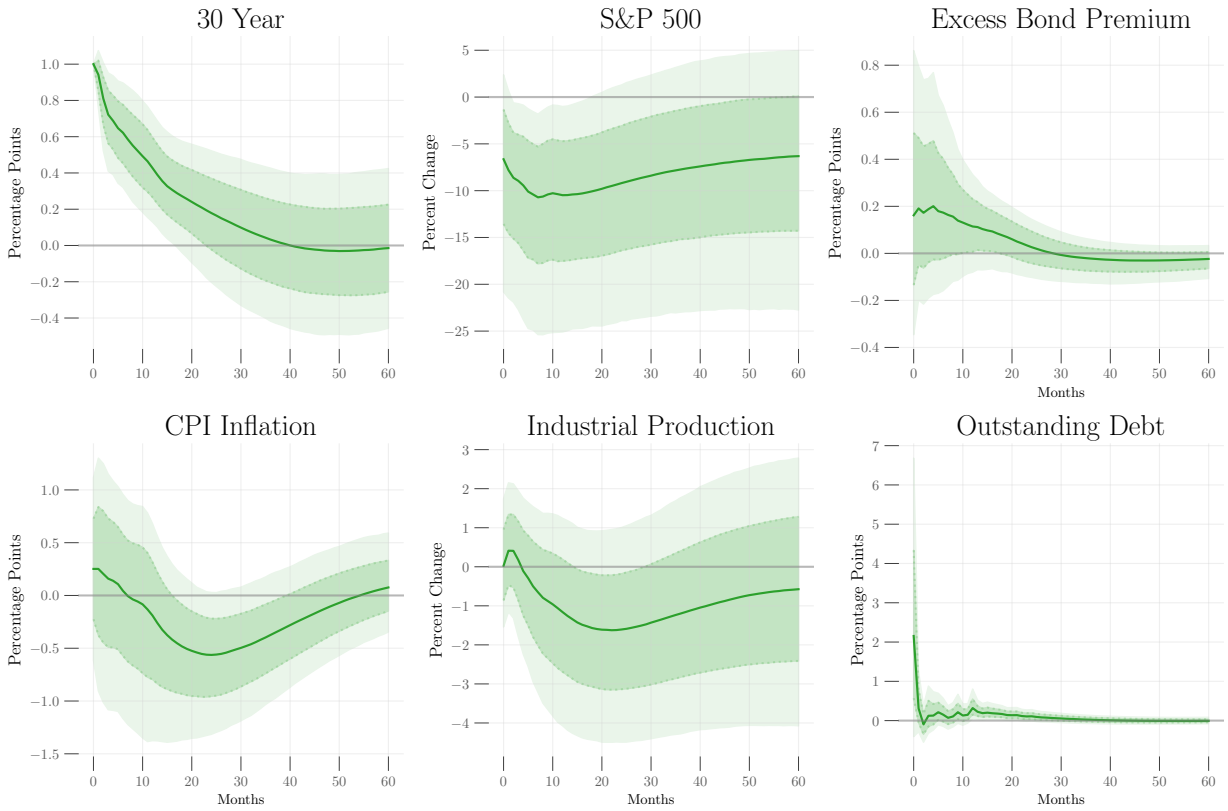


Figure A4: IRFs to an Identified Debt Shock – 30 Minute Windows

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set. Robust Wald statistic: 20.3 (full reduced-form sample: 10.4).

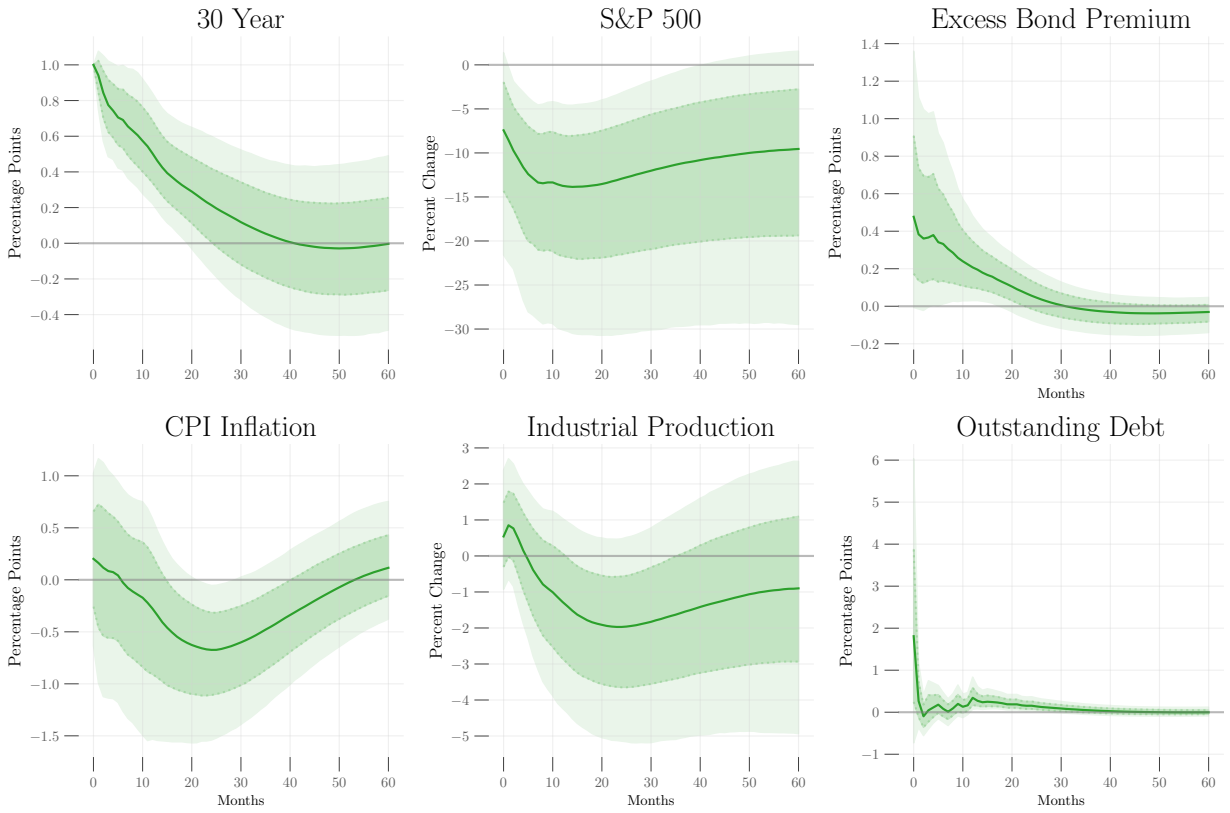


Figure A5: IRFs to an Identified Debt Shock – 60 Minute Windows

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set. Robust Wald statistic: 25.5 (full reduced-form sample: 10.9).

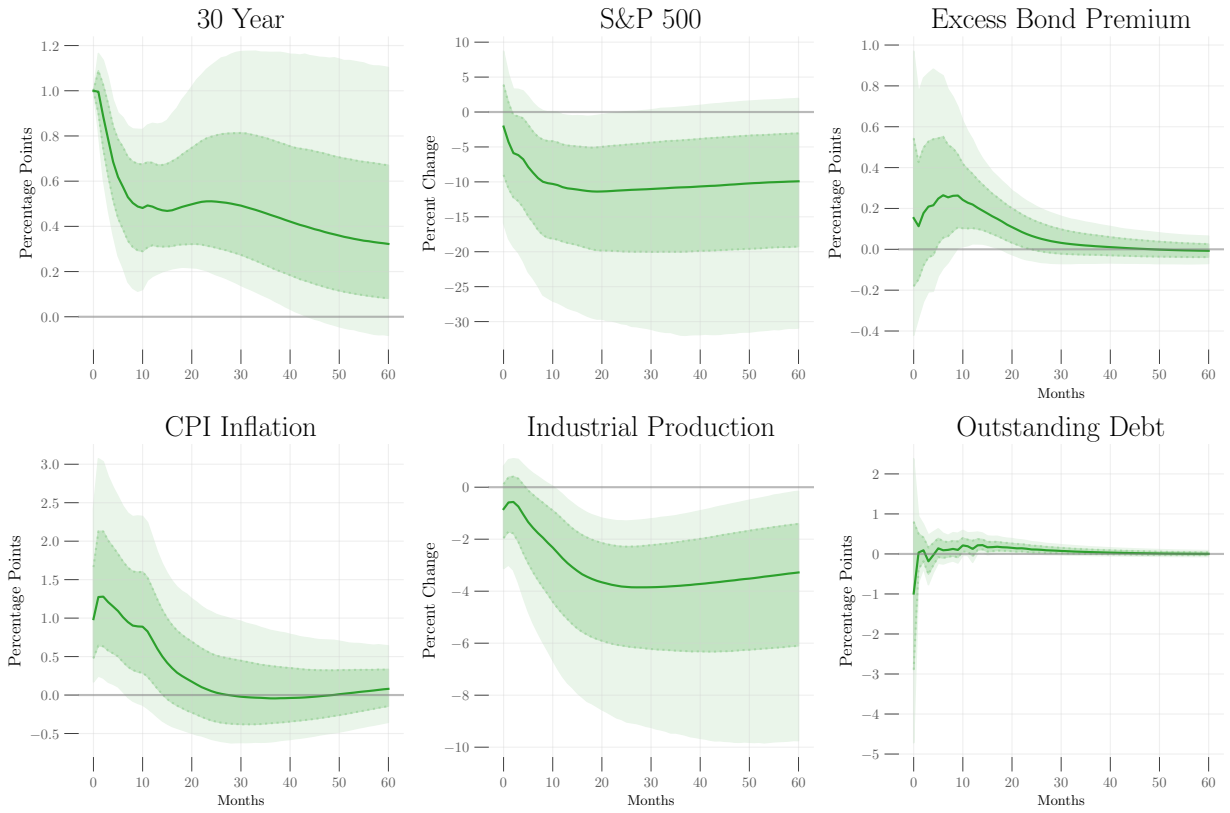


Figure A6: IRFs to an Identified Debt Shock – 5 Minute Windows

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set. Robust Wald statistic: 12.4 (full reduced-form sample: 7.9).

A6 Robustness to Lag Length

In my baseline specification, I choose 12 lags. Here, I first increase the number of lags to 24. I then decrease the number of lags to two as suggested by the BIC. The impulse responses are essentially unchanged.

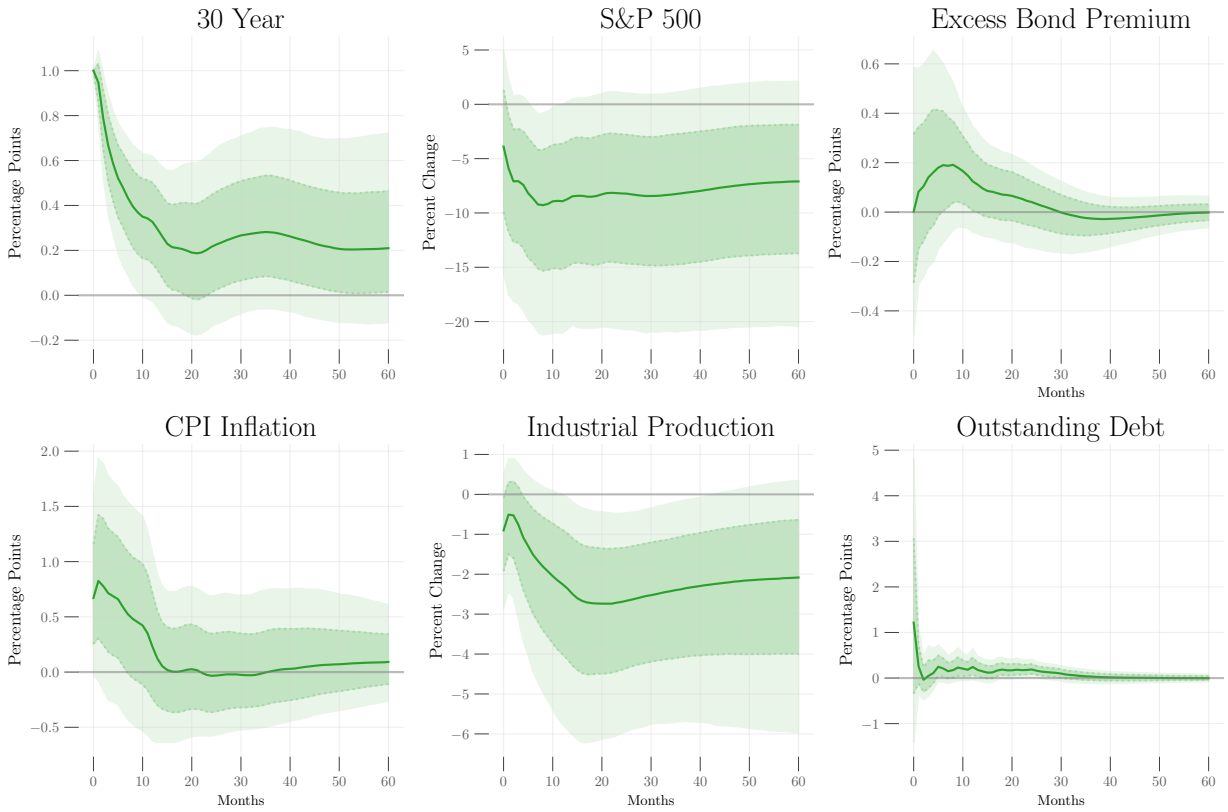


Figure A7: IRFs to an Identified Debt Shock – 24 lags

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set. Robust Wald statistic: 33.9 (full reduced-form sample: 9.9).

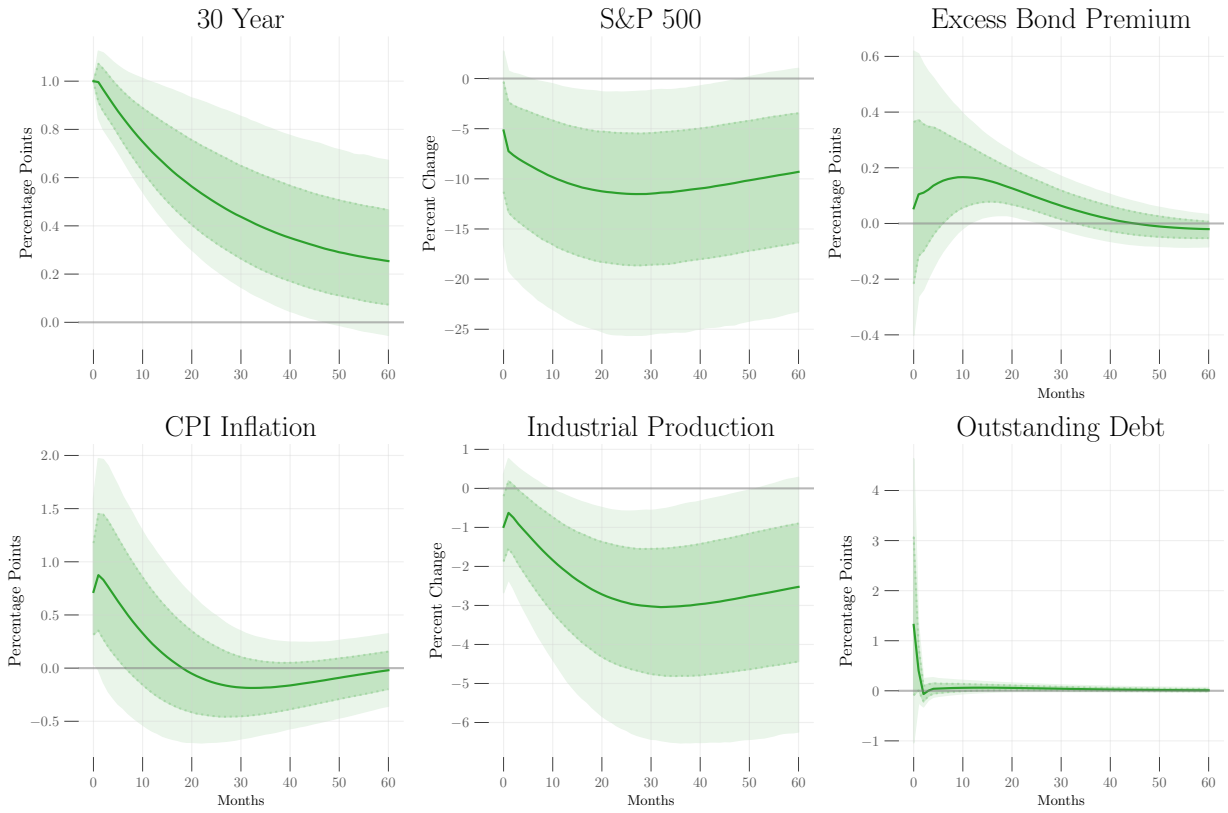


Figure A8: IRFs to an Identified Debt Shock – 2 lags

Note: Impulse responses are scaled such that the thirty-year yield increases by one percentage point. The solid green line denotes the median, the dark green area denotes the 68% credible set, and the light green area denotes the 90% credible set. Robust Wald statistic: 11.9 (full reduced-form sample: 8.4).

A7 Equilibrium Conditions

I list the equilibrium conditions for the New Keynesian model by sector.

Households

$$\omega_t = (C_t - hC_{t-1})^{-\gamma} - \beta h \mathbb{E}_t \left[(C_{t+1} - hC_t)^{-\gamma} \right] \tag{A1}$$

$$\omega_t = \psi_m (1-\rho_m)^{\xi} \left(M_t^{curr} - \rho_m M_{t-1}^{curr} \right)^{-\xi} + \beta \mathbb{E}_t \left[\Pi_{t+1}^{-1} \omega_{t+1} - \rho_m \psi_m (1-\rho_m)^{\xi} \left(M_{t+1}^{curr} - \rho_m M_t^{curr} \right)^{-\xi} \right] \tag{A2}$$

$$\Lambda_t = \beta \frac{\omega_t}{\omega_{t-1}} \tag{A3}$$

$$1 = \mathbb{E} \left[(1 + i_t^d) \Pi_{t+1}^{-1} \Lambda_{t+1} \right] \tag{A4}$$

$$C_t + M_t^{curr} = (1 + i_{t-1}^d) \Pi_t^{-1} d_{t-1} - d_t + w_t L_t + div_t - \tilde{X}_t - T_t + \Pi_t^{-1} M_{t-1}^{curr} \tag{A5}$$

Firms

$$w_t = \frac{1-\alpha}{1+\psi_1 \lambda_t} mc_t Z_t u_t^{\alpha} K_{t-1}^{\alpha} L_t^{\alpha} \tag{A6}$$

$$q_t = \mathbb{E}_t \left[\Lambda_{t+1} \left(q_{t+1} \left(1 - \delta(u_t) \right) + \alpha mc_{t+1} Z_{t+1} u_{t+1}^{\alpha} K_t^{\alpha - 1} L_{t+1}^{1-\alpha} \right) \right] \tag{A7}$$

$$1 + \lambda_t \psi_2 = q_t \left(1 - \frac{\partial S(I_t, I_{t-1})}{\partial I_t} I_t - S(I_t, I_{t-1}) \right) + \mathbb{E}_t \left[\Lambda_{t+1} q_{t+1} \frac{\partial S(I_{t+1}, I_t)}{\partial I_t} I_{t+1} \right] \tag{A8}$$

$$q_t \left(\frac{\partial \delta(u_t)}{\partial u_t} \right) K_{t-1} = \alpha mc_t Z_t u_t^{\alpha - 1} K_{t-1}^{\alpha} L_t^{1-\alpha} \tag{A9}$$

$$Q_t^L (1 + \lambda_t) = \mathbb{E}_t \left[(1 + \lambda_{t+1}) \Lambda_{t+1} Q_{t+1}^L \Pi_{t+1}^{-1} \kappa_L + \Lambda_{t+1} \Pi_{t+1}^{-1} \right] \tag{A10}$$

$$\psi_1 w_t L_t + \psi_2 I_t = Q_t^L \left(B_t^L - \kappa_L \Pi_t^{-1} B_{t-1}^L \right) \tag{A11}$$

$$K_t = I_t + (1 - \delta(u_t)) K_{t-1} - S(I_t, I_{t-1}) I_t \tag{A12}$$

$$Y_t = Z_t u_t^{\alpha} K_{t-1}^{\alpha} L_t^{1-\alpha} \tag{A13}$$

$$\ln \left(\Pi_t \Pi_{ss}^{-1} \right) = \psi_p \left(mc_t - \frac{\varepsilon_p - 1}{\varepsilon_p} \right) + \mathbb{E}_t \left[\Lambda_{t+1} \frac{Y_{t+1}}{Y_t} \ln \left(\Pi_{t+1} \Pi_{ss}^{-1} \right) \right] \tag{A14}$$

$$div_t = Y_t + Q_t^L (B_t^L - \kappa_L B_{t-1}^L \Pi_t^{-1}) - B_{t-1}^L \Pi_t^{-1} - w_t L_t - I_t - \frac{\varepsilon_p}{2\psi_p} \ln \left(\frac{\Pi_t}{\Pi_{ss}} \right)^2 Y_t \tag{A15}$$

Labor Unions

$$\Pi_t^w = \Pi_t \frac{w_t}{w_{t-1}} \tag{A16}$$

$$\ln(\Pi_t^w \Pi_{ss}^{-1}) = \psi_w \left(\chi L_t^{1+\eta} - \frac{\varepsilon_w - 1}{\varepsilon_w} \omega_t w_t L_t \right) + \beta \mathbb{E}_t \left[\ln(\Pi_{t+1}^w \Pi_{ss}^{-1}) \right] \tag{A17}$$

Financial Intermediaries

$$d_t + n_t = Q_t^L B_t^L + Q_t^B B_t^P + M_t^{res} \tag{A18}$$

$$1 + i_t^L = \frac{1 + \kappa_L Q_t^L}{Q_{t-1}^L} \tag{A19}$$

$$1 + i_t^B = \frac{1 + \kappa_B Q_t^B}{Q_{t-1}^B} \tag{A20}$$

$$\Omega_t = 1 - \sigma + \sigma \phi_t \theta \tag{A21}$$

$$\phi_t n_t = Q_t^L B_t^L + \Delta Q_t^B B_t^P \tag{A22}$$

$$\phi_t = \frac{\mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (1 + i_t^d) \right]}{\mathbb{E}_t \left[\theta - \Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_{t+1}^L - i_t^d) \right]} \tag{A23}$$

$$\frac{\mu_t}{1 + \mu_t} \Delta\theta = \mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_{t+1}^B - i_t^d) \right] \tag{A24}$$

$$\frac{\mu_t}{1 + \mu_t} \theta = \mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_{t+1}^L - i_t^d) \right] \tag{A25}$$

$$0 = \mathbb{E}_t \left[\Lambda_{t+1} \Omega_{t+1} \Pi_{t+1}^{-1} (i_t^r - i_t^d) \right] \tag{A26}$$

$$n_t = \frac{\sigma}{\Pi_t} \Big[(i_t^B - i_{t-1}^d) Q_{t-1}^B B_{t-1}^P + (i_t^L - i_{t-1}^d) Q_{t-1}^L B_{t-1}^L + (i_{t-1}^r - i_{t-1}^d) M_{t-1}^{res} + (1 + i_{t-1}^d) n_{t-1} \Big] \tag{A27}$$

$$\tilde{X}_t = \frac{n_t - X}{\sigma} \tag{A28}$$

Central Bank

$$Q_t^B B_t^{cb} = M_t^{res} + M_t^{curr} + TGA_t \tag{A29}$$

$$T_t^{cb} + Q_t^B B_t^{cb} = (1 + \kappa_B Q_t^B) \Pi_t^{-1} B_{t-1}^{cb} + M_t^{curr} + M_t^{res} + TGA_t - \Pi_t^{-1} M_{t-1}^{curr} - (1 + i_{t-1}^{res}) \Pi_t^{-1} M_{t-1}^{res} - \Pi_t^{-1} TGA_{t-1} \tag{A30}$$

$$i_t^{res} = (1 - \rho_r) i_{ss}^{res} + \rho_r i_{t-1}^{res} + (1 - \rho_r) \phi_\pi \left[\ln(\Pi_t) - \ln(\Pi_{ss}) \right] \tag{A31}$$

Government

$$B_t = B_t^P + B_t^{cb} \tag{A32}$$

$$Q_t^B B_t + T_t^{cb} = (1 + \kappa_B Q_t^B)\Pi_t^{-1} B_{t-1} + TGA_t + G_t - T_t - \Pi_t^{-1} TGA_{t-1} \tag{A33}$$

$$\ln(B_t) = (1 - \rho_b)\ln(B_{ss}) + \rho_b\left[\ln(B_{t-1}) + \ln(\Pi_{ss}) - \ln(\Pi_t)\right] + \varepsilon_t^b \tag{A34}$$

$$\ln\left(\frac{TGA_t}{TGA_{ss}}\right) = \rho_{tga}\ln\left(\frac{TGA_{t-1}}{TGA_{ss}}\right) + \gamma_{tga}\frac{Q_{ss}^B B_{ss}\varepsilon_t^b}{TGA_{ss}} \tag{A35}$$

Aggregation

$$Y_t = C_t + I_t + G_t + \frac{\varepsilon_p}{2\psi_p}\ln\left(\frac{\Pi_t}{\Pi_{ss}}\right)^2 Y_t \tag{A36}$$

Altogether, there are 35 equations with 35 unknowns. The aggregate resource constraint (equation A36) automatically clears due to Walrasian equilibrium. Note that for parsimony, I enforce $\psi_1 = \psi_2$.

A8 Calibration

Table A3
Calibrated Values

Parameter	Description	Target
M	Steady state monetary base	$\frac{M}{12Y} \approx 0.20$
M^{curr}	Steady state currency	$\frac{M^{curr}}{12Y} \approx 0.07$
TGA	Treasury General Account	$\frac{\text{TGA}}{12Y} \approx 0.025$
B	Steady state government debt	$\frac{Q^B B}{12Y} \approx 1.10$
Y	Output	1.00
T	Taxes	$\Pi \approx 2\%$
G	Government spending	0.20
i^b	Steady state government bond rate	$1.03^{1/12} - 1$
i^r	Steady state reserve rate	$1.02005^{1/12} - 1$
i^d	Steady state deposit rate	$1.02005^{1/12} - 1$
ε_p	Price elasticity	11
Z	Steady state productivity	1.00
κ_L	Firm loan decay	$1 - \frac{1}{60}$
κ_B	Government bond decay	$1 - \frac{1}{60}$
Δ	Spread on firm loans	$\frac{1}{2}$
γ	Consumption curvature	1.00
ξ	Currency curvature	1.00
η	Inverse Frisch	0.50
σ	Bank survival probability	$0.95^{1/3}$
ϕ	Steady-state leverage	4.00
ρ_m	Household currency persistence	0.975
ε_w	Wage elasticity	11
h	Habit parameter	0.90
α	Capital share	0.33
ψ_1	Wage bill financing	0.81
ψ_2	Investment financing	0.81
δ_0	Constant depreciation	$\frac{0.025}{3}$
δ_2	Quadratic depreciation	$\frac{0.01}{3}$
ψ_p	Phillips curve slope	$\frac{1}{20}$
ρ_b	Debt persistence	0.99
ρ_r	Reserve rate persistence	0.90
ϕ_π	Monetary-policy response to inflation	1.50
γ_{TGA}	TGA response parameter	0.65
ρ_{TGA}	TGA persistence	0.98
ψ_w	Wage Phillips curve slope	$\frac{1}{20}$
κ_I	Investment adjustment parameter	45

A9 Impulse Responses under Parameter Changes

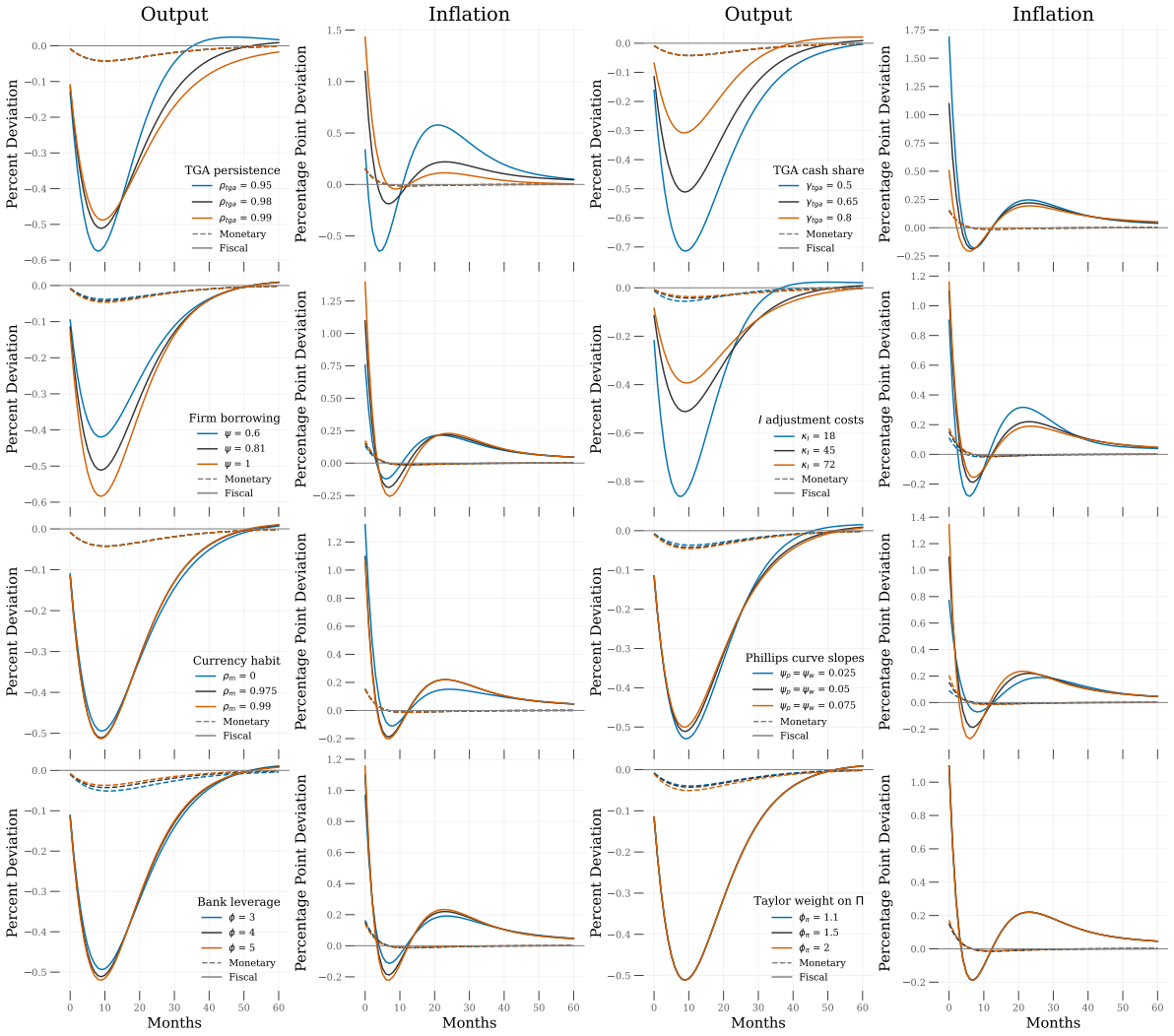


Figure A9: Impulse Response Functions to a 1.25 percent government debt increase in the financial New Keynesian model under alternative parameterization. Dashed lines represent monetary dominance, solid lines represent fiscal dominance.

A10 Simple Money-in-Utility Model

The equilibrium equations for the textbook New Keynesian money-in-utility model from Section 4 are as follows:

$$C_t^{-\gamma} = \beta \mathbb{E}_t \left[(1 + i_t) \Pi_{t+1}^{-1} C_{t+1}^{-\gamma} \right] \tag{A37}$$

$$\Lambda_t = \beta \frac{C_t^{-\gamma}}{C_{t-1}^{-\gamma}} \tag{A38}$$

$$\chi L_t^{\eta} = w_t C_t^{-\gamma} \tag{A39}$$

$$\psi_m M_t^{-\xi} = C_t^{-\gamma} \frac{i_t}{1 + i_t} \tag{A40}$$

$$Y_t = Z_t L_t \tag{A41}$$

$$w_t = Z_t mc_t \tag{A42}$$

$$\ln(\Pi_t) = \psi_p \left(mc_t - \frac{\varepsilon_p - 1}{\varepsilon_p} \right) + \mathbb{E}_t \left[\Lambda_{t+1} \frac{Y_{t+1}}{Y_t} \ln(\Pi_{t+1}) \right] \tag{A43}$$

$$B_t + T_t + M_t = G_t + (1 + i_{t-1}) \Pi_t^{-1} B_{t-1} + \Pi_t^{-1} M_{t-1} \tag{A44}$$

$$\ln(B_t) = (1 - \rho_b) \ln(B_{ss}) + \rho_b \left[\ln(B_{t-1}) - \ln(\Pi_t) \right] + \varepsilon_t^b \tag{A45}$$

$$i_t = (1 - \rho_r) i_{ss} + \rho_r i_{t-1} + (1 - \rho_r) \phi_\pi \ln(\Pi_t) \tag{A46}$$

$$Y_t = C_t + G_t + \frac{\varepsilon_p}{2\psi_p} \ln \left(\frac{\Pi_t}{\Pi_{ss}} \right)^2 Y_t \tag{A47}$$